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Cardiovascular Risk Prediction and Longitudinal Lifestyle Change in UK Biobank: Predictive and Causal Machine-Learning Models

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1. ABSTRACT

Background. Cardiovascular disease remains a leading cause of death worldwide. Established risk tools identify high-risk individuals but provide limited insight into whether lifestyle changes translate into meaningful risk reductions.

Methods. Using UK Biobank, causal machine learning estimated effects of lifestyle change on incident cardiovascular disease. In 71,428 participants with six lifestyle domains (sleep, diet, alcohol, smoking, physical activity, mental health) measured at baseline and imaging visits, doubly robust causal forests estimated average treatment effects for 24 single-domain transitions and multi-arm forests estimated 120 joint contrasts. Sensitivity analyses addressed outcome timing, treatment-group overlap, and unmeasured confounding. A complementary baseline prediction model, comparing logistic regression with three tree-based ensembles in 458,840 participants, supplied absolute CVD risk. Causal estimates were externally validated in NHEFS, and both streams were integrated into an interactive tool.

Results. All four prediction models achieved good discrimination (AUC-ROC 0.714-0.718) with excellent calibration, while logistic regression was retained for interpretability. Most lifestyle transitions showed no evidence of effect, and well-supported transitions excluded absolute risk differences exceeding approximately one percentage point. Mental health was the only domain showing directionally coherent associations. The single transition surviving multiple-testing correction did not withstand temporal and confounding scrutiny. External validation in NHEFS showed directional agreement for eight of eleven testable contrasts with overlapping confidence intervals throughout. The interactive tool was implemented as a working prototype presenting predicted risk and lifestyle-change estimates with their uncertainty.

Conclusions. Lifestyle and demographic information alone supported good CVD risk discrimination without clinical measurements. This study integrates baseline risk prediction, timing-aware causal analysis of longitudinal lifestyle change, external validation and interactive risk communication within a single framework. The timing-aware design identified a superficially robust finding as reverse causation, and narrow confidence intervals converted null results into informative bounds. Future studies

135 with repeated assessments, longer follow-up, and richer covariates can build directly on this framework.

Keywords: Cardiovascular Disease | Lifestyle Change | Longitudinal Data | Risk Prediction
| Causal Forests | Causal Inference | UK Biobank | Machine Learning

2. INTRODUCTION

Cardiovascular disease (CVD) remains a leading cause of mortality and morbidity worldwide, accounting for approximately 17.9 million deaths each year and 32% of all global deaths [1, 2]. In the United Kingdom, more than eight million people are living with CVD, and this number is expected to increase as the population ages [3].

CVD comprises a broad group of disorders of the heart and blood vessels, with ischaemic heart disease, stroke and heart failure among the principal causes of CVD-related morbidity and mortality. CVD typically develops over decades before culminating in acute events such as myocardial infarction or stroke, creating a long preclinical window in which early risk prediction and sustained prevention may be beneficial [1].

Risk of CVD is determined not only by genetic predisposition and clinical risk factors, such as elevated blood pressure, adverse blood lipids and diabetes, but also by modifiable lifestyle factors, including smoking, alcohol consumption, sleep, diet, physical activity and mental health. Healthier lifestyle patterns have consistently been associated with lower risks of CVD and all-cause mortality, so identifying high-risk individuals and understanding modifiable lifestyle factors are both important for reducing this burden [4, 5, 6].

Several clinical risk prediction tools support the early identification of individuals at increased cardiovascular risk. SCORE2 was derived from European cohorts [7], QRISK3 is widely used in the United Kingdom [8], and the Pooled Cohort Equations underlie the ASCVD risk estimator recommended in the United States [9]. These models combine established demographic and clinical predictors such as age, sex, smoking status, blood pressure and lipids, but represent broader lifestyle behaviours only to a limited extent and generally at a single time point. It therefore remains unclear how well comprehensive lifestyle profiles support CVD risk prediction and how longitudinal changes in these behaviours relate to subsequent risk [10, 11].

Large prospective cohorts provide an opportunity to integrate detailed demographic and lifestyle information with longitudinal health outcomes. Machine-learning methods can accommodate nonlinear associations and interactions that may not be captured by conventional models. Greater model complexity does not necessarily improve prediction, so that any improvement must be demonstrated through appropriate validation [11, 12,

13, 1]. However, predictive performance does not establish whether changing a lifestyle
behaviour would alter subsequent CVD risk.

Estimating the effect of lifestyle change requires a causal rather than purely predictive
framework. Prediction models estimate risk under observed conditions, whereas causal
inference considers how outcomes might differ under alternative interventions. In
observational data, this requires explicit assumptions concerning treatment definition,
confounding, temporal ordering and outcome ascertainment, particularly when
randomised trials are unavailable or impractical.

Causal machine learning combines these principles with flexible methods for estimating
treatment-effect variation. Causal forests estimate both average and conditional average
treatment effects, allowing the effects associated with lifestyle change to be examined across
characteristics such as age and BMI. These estimates represent average effects among
participants with similar measured characteristics rather than directly observed individual
effects. Therefore, their interpretation depends on adequate control of confounding,
sufficient treatment-group overlap and careful assessment of temporal ordering. [14, 15].

The UK Biobank provides a setting in which predictive and causal analyses can be
conducted within the same prospective cohort. Its large baseline sample supports
risk-prediction modelling, while repeated lifestyle assessments and linked health records
allow behavioural change and subsequent cardiovascular outcomes to be examined
longitudinally [16, 17]. However, some cardiovascular events occurred before the second
lifestyle assessment, creating temporal ambiguity and a potential for reverse causation.

Previous studies have extensively examined CVD prediction and lifestyle-associated risk,
but few have integrated baseline prediction with causal analyses of longitudinal lifestyle
change within the same cohort. The timing of cardiovascular events relative to repeated
lifestyle assessment has also received limited attention. Here, we compare conventional and
machine-learning models for baseline CVD prediction and use causal forests to estimate
risk differences associated with individual and combined lifestyle changes. Temporal
sensitivity analyses, unmeasured confounding analyses and baseline lifestyle comparisons
are used to assess the credibility of these estimates.

The specific objectives are:

- To construct and internally validate a baseline model for predicting incident CVD risk, comparing conventional logistic regression with tree-based ensemble machine-learning methods.
- To assess the predictive value of self-reported lifestyle and basic demographic information alone, benchmarked against the published performance of established clinical risk calculators.
- To estimate the average and heterogeneous effects of changes in individual lifestyle factors between the baseline and imaging visits using causal-forest methods, explicitly considering whether CVD events occurred before or after the second measurement.
- To explore the joint effects of combined lifestyle changes using multi-arm causal forests where sufficient treatment-group overlap is available.
- To estimate the effects of baseline lifestyle levels as a temporally unambiguous reference for interpreting the change estimates.
- To develop an interactive research prototype that allows users to explore predicted CVD risk and observational estimates associated with alternative lifestyle scenarios.

By integrating risk prediction, causal inference, multi-exposure modelling and longitudinal lifestyle information within a timing-aware framework, this study seeks to bridge the gap between identifying who is at increased CVD risk and understanding which modifiable factors may contribute to it, treating prediction and causation not as interchangeable but as complementary components of personalised CVD prevention research.

3. METHODS

3.1 Study design and data source

The present study was a retrospective analysis of prospectively collected data from the UK Biobank (UKB), a large population-based cohort established to investigate genetic, environmental and lifestyle determinants of disease [16]. Between 2006 and 2010, approximately 500,000 participants aged 40–69 years were recruited through 22 assessment centres across England, Scotland and Wales. Data were collected via touchscreen questionnaires, interviews, physical measurements and biological samples, and a subset returned for an imaging visit from 2014 onwards, at which the lifestyle measures were repeated [18, 16, 19].

3.2 Study population

All individuals in the baseline survey ($n = 502,510$) were considered, and those with CVD recorded before baseline were excluded, leaving 459,142 participants. CVD was defined from linked hospital and mortality records [20]. A further 302 records ($< 0.1\%$) attended the imaging visit but had one or more lifestyle domains that could not be scored at that visit. To keep the causal cohort a consistent subset of the predictive cohort, these records were removed from both rather than imputed, acknowledging possible selection bias and giving a final baseline cohort of 458,840. The longitudinal causal cohort comprised 71,428 participants with eligible lifestyle measurements at both the baseline and imaging visits. The flow of inclusion and exclusion is shown in Figure 3.1. The clinical characteristics of both cohorts, including incident event counts, are presented in Table S1.1.

3.3 Assessment of lifestyle variables

Six modifiable lifestyle domains were assessed: sleep, diet, alcohol consumption, smoking, physical activity, and mental health. Definitions, coding rules and score directions were identical at both assessments. Each domain was summarised as a three-level ordered categorical score with constituent items contributing evenly. Higher scores indicate healthier behaviour for sleep, diet and physical activity, and greater burden for alcohol,

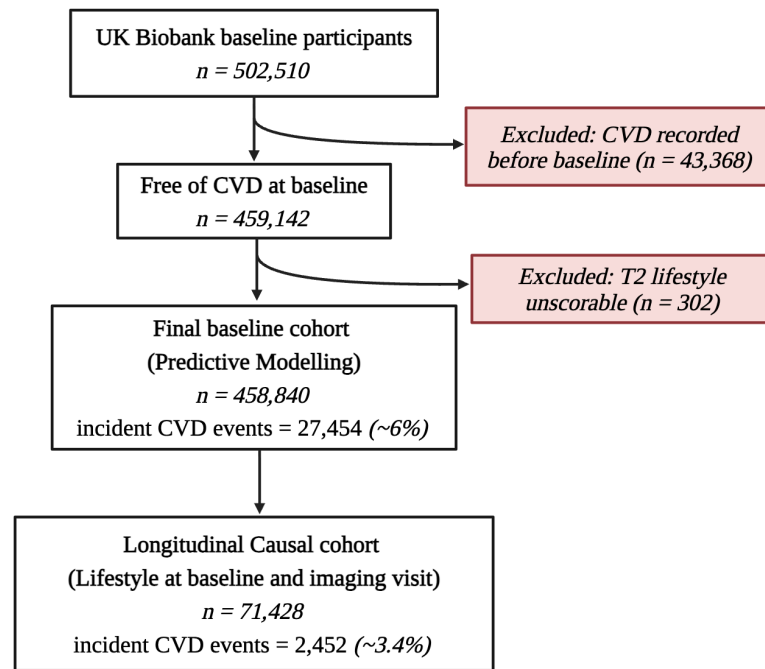


Figure 3.1: Participant selection for the predictive cohort and longitudinal causal cohort.

smoking and mental health. In brief, sleep was assessed from sleep duration, chronotype, insomnia, snoring and daytime dozing [19], and diet captured adherence to American Heart Association food-group targets [21, 22, 23]. Alcohol combined drinking status with intake frequency, with the three levels corresponding to never or former drinking, occasional drinking, and drinking at least weekly [24]. Smoking was classified as never, former or current [25]. Physical activity was derived from the short-form International Physical Activity Questionnaire as metabolic equivalent of task (MET)-weighted volume in three levels [26, 27, 28], and mental health combined the two-item Patient Health Questionnaire (PHQ-2), the two-item Generalised Anxiety Disorder scale (GAD-2) and loneliness [19]. Exact UKB field identifiers, thresholds, missing-value rules and score construction are in the Supplementary S1.1.

3.4 Predictive modelling

Participants and outcome

The predictive analysis used nine predictors, namely age, sex, body mass index (BMI) and the six lifestyle scores, together with the outcome. The primary outcome was incident

CVD after baseline ($N = 458,840$, prevalence 5.98%). A broader composite of CVD, atrial fibrillation or heart failure served as a secondary outcome, reflecting the contemporary view of CVD as a cardiometabolic disease that extends beyond ischaemic and atherosclerotic conditions to include atrial fibrillation and heart failure. The identical pipeline was applied to both outcomes, so any difference is attributable to the outcome definition alone (Table 3.1).

Table 3.1: Cohort summary

Outcome	N	Events	Event rate (%)	Train N	Test N
CVD	458,840	27,454	5.98	321,188	137,652
CVD/HF/AF	453,658	37,016	8.16	317,560	136,098

Note: CVD, cardiovascular disease; HF, heart failure; AF, atrial fibrillation.

Data preprocessing

Participants were randomly divided into training (70%) and testing (30%) sets, stratified to preserve outcome prevalence. Missingness was below 3% for every predictor. Continuous predictors (age, BMI and the alcohol, diet and mental-health scores) were imputed with iterative chained-equations imputation [29] over 10 cycles, yielding a single completed dataset, and categorical predictors (sex and the sleep, smoking and physical-activity categories) were assigned their most frequent category. All imputation models were fitted on the training set only and applied unchanged to the testing set, preventing information leakage. For logistic regression, lifestyle scores were standardised (zero mean, unit variance) after imputation, and cubic splines with four knots were applied to age and BMI to capture non-linear associations. Tree-based models used untransformed inputs, as they are invariant to monotonic transformations.

Model development

Four algorithms were evaluated: penalised logistic regression, Random Forest [30], XGBoost [31] and LightGBM [32]. Logistic regression was treated as the primary model given its clinical interpretability and coefficient stability, while the tree-based methods assessed whether modelling non-linear relationships and higher-order interactions

improved performance. Hyperparameters were tuned on the training set only, using stratified cross-validation optimising the area under the receiver operating characteristic (ROC) curve (search strategies, grids and settings in Supplementary Table S1.2 and Methods S1.2). The best configurations were refitted on the full training set before final evaluation on the held-out test set.

Model evaluation

All performance metrics were computed on the held-out test set. Discrimination was assessed by the area under the AUC-ROC with 95% confidence intervals (CI) from 1,000 bootstrap replicates, and compared between models with the DeLong test. Precision–recall curves were additionally reported given class imbalance. Calibration was evaluated by the calibration slope, calibration-in-the-large and the Brier score with bootstrap confidence intervals, and clinical utility by decision curve analysis across threshold probabilities of 1–50% against treat-all and treat-none strategies. Performance was stratified by sex and age (< 60 vs ≥ 60 years), with stratum sizes and events in Table S2.4. Feature importance was summarised by odds ratios (logistic regression) and Gini/gain-based scores (tree models).

3.5 Causal machine learning

Estimand specification

Causal analyses estimated the effect of longitudinal lifestyle change on incident CVD, using a target-trial-emulation framework to define the target population, treatment strategies, comparisons, outcome window, adjustment set and estimands before model fitting. The target population comprised UKB participants with complete lifestyle assessments at baseline (T0) and the imaging visit (T2), no prevalent CVD at T0, and available outcome data over follow-up (Figure 3.2). Treatment was defined as the change in each lifestyle variable between T0 and T2, operationalised as ordinal category transitions classified as improvement, deterioration or remaining stable. Participants who changed category (“movers”) were compared with those who remained in their baseline category at both time points (“stayers”); conditioning both arms on a common baseline category prevents

baseline level from confounding the effect of change.

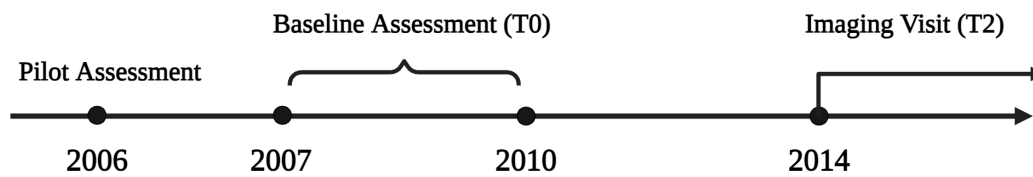


Figure 3.2: Timeline of the pilot, baseline, and imaging assessments used in the longitudinal analysis.

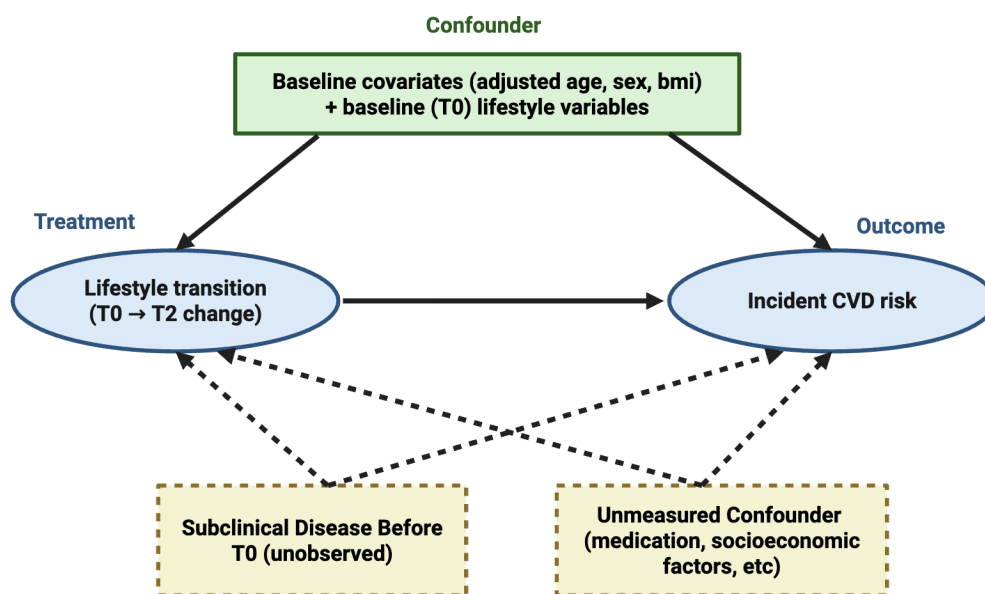


Figure 3.3: Directed acyclic graph relating lifestyle transition to incident CVD risk, with measured and potential unmeasured confounders.

The assumed causal structure was encoded in a pre-specified directed acyclic graph (DAG; Figure 3.3). Baseline covariates (age, sex, BMI and T0 lifestyle levels) are common causes of the transition and incident CVD, opening back-door paths that the adjustment set closes, whereas post-baseline variables (T2 lifestyle levels, interim health events) lie on mediating or collider paths and were excluded. The DAG makes two residual threats explicit: unmeasured confounding, addressed through E-values and quantitative bias analysis, and reverse causation from subclinical disease to lifestyle change, addressed through the strict-temporal sensitivity analysis and the baseline reference model below.

The primary outcome was incident CVD diagnosed from T0 through the end of follow-up, defined as up to 10 years and censored at the event date, death or the end of record

linkage, a broad window capturing all events regardless of timing relative to T2. A stricter definition excluding events that occurred before T2 was examined in sensitivity analysis.

325 Consistent with the DAG, the adjustment set comprised age, sex, BMI and baseline (T0) values of the five other lifestyle variables (the domain under study defines the transition and was excluded from its own set).

Both analyses target the same estimand. With treatment arm k compared against the stable reference arm 0, the conditional average treatment effect is

$$\tau_k(X_i) = \mathbb{E}[Y_i(k) - Y_i(0) \mid X_i]$$

330 where $Y_i(k)$ and $Y_i(0)$ are the potential CVD outcomes under changing and staying and X_i is the adjustment set above; k is a single directional transition in the single-variable analysis and indexes the joint-change arms in the combined analysis. The reported average treatment effect (ATE) for each arm is the population average $\mathbb{E}[\tau_k(X)]$.

Baseline causal reference model

335 A baseline-level causal reference analysis in the baseline cohort compared healthy with less healthy lifestyle categories before outcome occurrence (Supplementary S1.3).

Single-variable analysis

For each of the six lifestyle domains, four directional transitions were defined, giving 24 contrasts, each estimated within the sub-cohort sharing the relevant baseline category.

340 No transition was excluded by a sample-size or event-count threshold. Arm size, events, overlap and estimability were reported for every contrast, and contrasts with fewer than 50 events in the smaller arm were labelled thin support, retained but interpreted cautiously.

Each transition was estimated with a binary causal forest fitted in the grf package [33, 34], which estimates conditional average treatment effects (CATEs) non-parametrically

345 with honest sample splitting and local centering. Forests used 5,000 trees with cross-validated tuning. Individual-level CATEs were taken from the fitted forests, and the overall ATE from grf's *average_treatment_effect* function, which averages doubly robust scores

rather than predicted CATEs. Estimates are reported as percentage point differences in CVD risk with asymptotic standard errors, Wald 95% CI and two-sided p-values.

Benjamini–Hochberg adjustment [35] was applied across the 24 transitions, classifying findings as FDR-supported, nominally significant only, or showing no evidence of an effect.

Covariate balance was assessed by standardised mean differences (SMDs) before and after overlap weighting, and positivity by propensity-score distributions and effective sample sizes. Subgroup-specific ATEs were estimated by sex, age (< 60 vs ≥ 60 years) and BMI category, with differences tested by bootstrap of the doubly robust scores and best linear projection (BLP) interaction tests. CATE distributions, calibration tests and variable-importance measures were treated as supportive or exploratory analyses. Rank-average treatment effect (RATE) statistics were computed but excluded as inadmissible because participants were ranked on the forests’ own out-of-bag predictions (Supplementary S1.5).

Combined-variable analysis

Joint effects of two lifestyle changes were examined across the 15 pairwise combinations of the six domains, with multi-arm treatment groups representing improvement, deterioration or mixed-change scenarios. Participants stable in both domains formed the pre-specified reference group. A structural-feasibility filter restricted each analysis to participants whose baseline category allowed movement in the required direction, so the reference group excluded participants unable to change. All feasible arms were retained with the same support diagnostics.

Multi-arm causal forests (grf, 3,000 trees, with the same honest sample splitting and local centering) estimated the estimand defined above for each active arm relative to the stable reference. As cross-validated tuning is not available for the multi-arm implementation, default grf settings were used. Joint-treatment effects are reported with asymptotic confidence intervals and FDR-adjusted p-values. Departure from additivity compared the observed joint effect with the sum of the individual effects, without bootstrap inference for the primary multi-arm forests. These interaction estimates are descriptive rather than formal evidence of synergy.

Sensitivity analyses

Sensitivity analyses addressed temporal ambiguity, overlap and unmeasured confounding. First, all models were repeated using only post-T2 CVD events, so the lifestyle change strictly preceded the outcome. Second, full-sample, overlap-weighted and propensity-trimmed estimates were compared under progressively stricter trimming rules, reporting
380 the retained sample fraction under each. Third, E-values and quantitative bias analysis (QBA) quantified the unmeasured confounding required to explain away the observed associations (Supplementary Methods S1.7).

Internal validation

385 Internal validity was supported by three procedures. First, honest estimation was used within each forest, reducing overfitting and supporting valid statistical inference. Second, predicted CATEs were evaluated for calibration by dividing participants into quartiles of predicted effect, independently estimating the ATE within each quartile, and comparing these estimates with the corresponding mean predicted CATEs. Third, a fixed random
390 seed was used throughout the primary analysis to ensure computational reproducibility (Supplementary Methods S1.4~S1.6).

3.6 External validation

External validation was conducted in the NHANES I Epidemiologic Follow-up Study (NHEFS), a US population-based cohort[36]. The temporal design mirrored the primary
395 analysis: lifestyle and covariates were measured in 1982–84 (T0), lifestyle change in 1987 (T2), and incident CVD from hospitalisation and mortality records during 1988–92. Applying equivalent eligibility criteria yielded 7,446 participants and 629 primary events (8.4%; ischaemic heart disease or stroke; ICD-9 410–414 and 430–438). A broader outcome additionally included heart failure and atrial fibrillation (751 events; 10.1%). Lifestyle
400 measures were harmonised to the UK Biobank ordinal scales, with details and limitations reported in the Supplementary Methods S1.8. Because repeated measures were available only for smoking, alcohol and physical activity, validation was restricted to transitions in

these domains and their pairwise joint contrasts; the remaining domains were included only as baseline covariates. The same causal-forest pipeline was applied across three random
405 seeds. Given differences in population, period and measurement, validation focused on agreement in effect direction and uncertainty rather than equality of effect sizes.

3.7 Visualisation tool

The two analytic streams were combined in an interactive risk simulator built with Streamlit in Python, framed as a research tool rather than a clinical decision system. For a
410 user-specified profile, the prediction model supplies the absolute baseline CVD risk and the causal-forest estimates supply the risk difference for a proposed lifestyle transition among similar participants, both displayed with confidence intervals and observational-design caveats.

4. RESULTS

4.1 Predictive modelling

Discrimination

All four models achieved similar discrimination on the test set (Table 4.1). For the primary CVD outcome, logistic regression and XGBoost obtained the highest AUC (0.718, 95% CI 0.712–0.724), followed by LightGBM (0.715) and random forest (0.714) (Figure S2.1). Average precision was much lower (approximately 0.13), so precision–recall curves better reflect performance in this imbalanced setting (Figure S2.2). DeLong tests showed that logistic regression significantly outperformed random forest and LightGBM but was statistically tied with XGBoost ($p = 0.45$), and the differences were clinically negligible (< 0.004 AUC units). The secondary outcome analysis produced comparable results (Table S2.1).

Table 4.1: Predictive-model performance for primary model

Model	AUC (95% CI)	Average precision	Brier score	Scaled Brier score	Calibration slope	Calibration intercept
Logistic regression	0.718 (0.712–0.724)	0.132	0.0542	0.036	0.984	−0.0001
Random Forest	0.714 (0.708–0.720)	0.129	0.0543	0.035	0.986	−0.001
XGBoost	0.718 (0.712–0.724)	0.132	0.0542	0.036	0.969	−0.0006
LightGBM	0.715 (0.709–0.720)	0.129	0.0543	0.034	0.922	0.0018

Calibration and overall performance

All models showed near-perfect calibration slopes (0.92 to 0.99), with logistic regression (0.984) and random forest (0.986) closest to 1.0, and calibration intercepts near zero, indicating negligible over- or under-prediction. Calibration plots showed predicted risks tracking observed event frequencies across deciles (Figure S2.3). Brier scores were low and similar across models (0.0542 to 0.0543), while the scaled Brier scores (0.034 to 0.036) indicated only modest improvement over the null model (Table 4.1). The secondary

outcome analysis showed comparable calibration with mild underprediction (Figure S2.3, Table S2.1).

Feature importance

Across all four models, age, sex and BMI were consistently the most influential predictors, and among the lifestyle domains smoking and mental health contributed most, with diet, sleep and physical activity contributing modestly. In the logistic regression model, male sex and current smoking were each associated with approximately twofold higher predicted odds of CVD relative to their reference categories. Full model-specific rankings and coefficients are reported in Supplementary Tables S2.2 and S2.3. These findings describe predictive associations and do not show that changing these variables would cause a change in CVD risk.

Subgroup performance and clinical utility

Discrimination was lower in older adults (age ≥ 60 , AUC-ROC 0.64 to 0.65) than in younger adults (age < 60 , 0.70 to 0.71), and was more comparable across sex (0.68 to 0.72, higher in women). The secondary outcome analysis showed the same pattern, indicating that the reduced performance in older adults is unlikely to be attributable to the outcome definition. Decision curve analysis showed no meaningful difference between models, with all four providing net benefit over the treat-all and treat-none strategies across threshold probabilities of approximately 6 to 17% (Figure S2.4).

Given practically equivalent discrimination, calibration and clinical utility, model selection was based on interpretability and computational simplicity: logistic regression is used as the primary model for CVD risk prediction in this cohort and supplies the absolute baseline risk for the interactive tool.

4.2 Causal machine learning

Causal cohort and transition support

The primary causal cohort comprised 71,428 participants with 2,452 incident CVD events (3.43%). The lower event rate than the baseline cohort reflects the healthier profile of participants attending the imaging visit. Arm sizes and events for the 24 pre-specified transitions are reported in Table 4.2. Five transitions with fewer than 50 events in the smaller arm are flagged as thin support and reported for completeness only. Of the 192 covariate-contrast SMDs across the eight adjustment covariates, 63 exceeded 0.1 before weighting and 2 after overlap weighting, so the weighted arms were well matched on measured covariates (Figure 4.1). Effective sample sizes under overlap weighting retained at least 95% of nominal arm sizes in 18 of 24 contrasts and at least 91% in all but one, including 92% for alcohol 2→1 contrast discussed below (Figure S3.1). The single poor case was smoking relapse, the transition from former to current smoking (64%), which indicates near-deterministic treatment assignment, and its estimates are treated as least reliable and revisited in the Discussion.

Table 4.2: Average treatment effects for the 24 pre-specified lifestyle transitions

Transition	Stayed	Moved	Events	ATE (pp)	95% CI	p	q Status
Sleep: improvement ($0 \rightarrow 1$)	7,186	6,799	589	-0.09	[-0.75, +0.57]	0.798	0.912 -
Sleep: deterioration ($1 \rightarrow 0$)	17,270	6,467	832	-0.02	[-0.54, +0.49]	0.926	0.926 -
Sleep: improvement ($1 \rightarrow 2$)	17,270	7,627	876	-0.13	[-0.61, +0.36]	0.606	0.818 -
Sleep: deterioration ($2 \rightarrow 1$)	14,787	11,292	762	0.16	[-0.25, +0.57]	0.441	0.662 -
Diet: improvement ($0 \rightarrow 1$)	12,738	11,245	931	0.07	[-0.42, +0.56]	0.788	0.912 -
Diet: deterioration ($1 \rightarrow 0$)	20,556	7,104	913	-0.19	[-0.65, +0.27]	0.426	0.662 -
Diet: improvement ($1 \rightarrow 2$)	20,556	6,682	898	0.52	[+0.01, +1.04]	0.047	0.187 Nominal (par.)
Diet: deterioration ($2 \rightarrow 1$)	6,517	6,586	369	0.63	[+0.06, +1.20]	0.031	0.148 Nominal (coh.)
PA: improvement ($0 \rightarrow 1$)	5,096	10,723	586	-0.14	[-0.76, +0.48]	0.659	0.832 -
PA: deterioration ($1 \rightarrow 0$)	14,694	3,668	638	0.36	[-0.32, +1.04]	0.296	0.635 -
PA: improvement ($1 \rightarrow 2$)	14,694	10,799	824	-0.11	[-0.55, +0.32]	0.613	0.818 -
PA: deterioration ($2 \rightarrow 1$)	17,401	9,047	893	0.23	[-0.22, +0.69]	0.318	0.635 -
Mental: deterioration ($0 \rightarrow 1$)	51,037	6,085	1,916	0.70	[+0.14, +1.25]	0.014	0.114 Nominal (coh.)
Mental: improvement ($1 \rightarrow 0$)	3,710	6,331	349	-0.95	[-1.71, -0.19]	0.014	0.114 Nominal (coh.)
Mental: deterioration ($1 \rightarrow 2$)	3,710	1,026	194	0.59	[-0.83, +2.01]	0.414	0.662 Thin
Mental: improvement ($2 \rightarrow 1$)	934	2,305	142	-1.22	[-2.97, +0.53]	0.172	0.412 Thin
Alcohol: deterioration ($0 \rightarrow 1$)	2,799	581	134	-1.38	[-2.93, +0.17]	0.080	0.214 Thin
Alcohol: improvement ($1 \rightarrow 0$)	9,179	1,422	367	1.04	[-0.07, +2.16]	0.065	0.196 -
Alcohol: deterioration ($1 \rightarrow 2$)	9,179	2,819	384	-0.34	[-1.07, +0.38]	0.354	0.653 -
Alcohol: improvement ($2 \rightarrow 1$)	47,660	6,968	1,868	0.98	[+0.45, +1.51]	0.0003	0.007 FDR (par.)
Smoking: deterioration ($0 \rightarrow 1$)	42,658	835	1,268	1.33	[-0.04, +2.70]	0.057	0.194 Thin
Smoking: not interpretable ($1 \rightarrow 0$)	20,725	2,100	950	0.06	[-0.87, +0.98]	0.906	0.926 N/I
Smoking: deterioration ($1 \rightarrow 2$)	20,725	428	883	0.15	[-2.28, +2.58]	0.902	0.926 Thin
Smoking: improvement ($2 \rightarrow 1$)	1,934	2,748	219	1.31	[+0.13, +2.50]	0.030	0.148 Nominal (par.)

Status: -, no evidence; Nominal, $p < 0.05$ but $q \geq 0.05$; FDR, FDR-supported; par., paradoxical; coh., coherent; Thin, thin support; N/I, not causally interpretable.

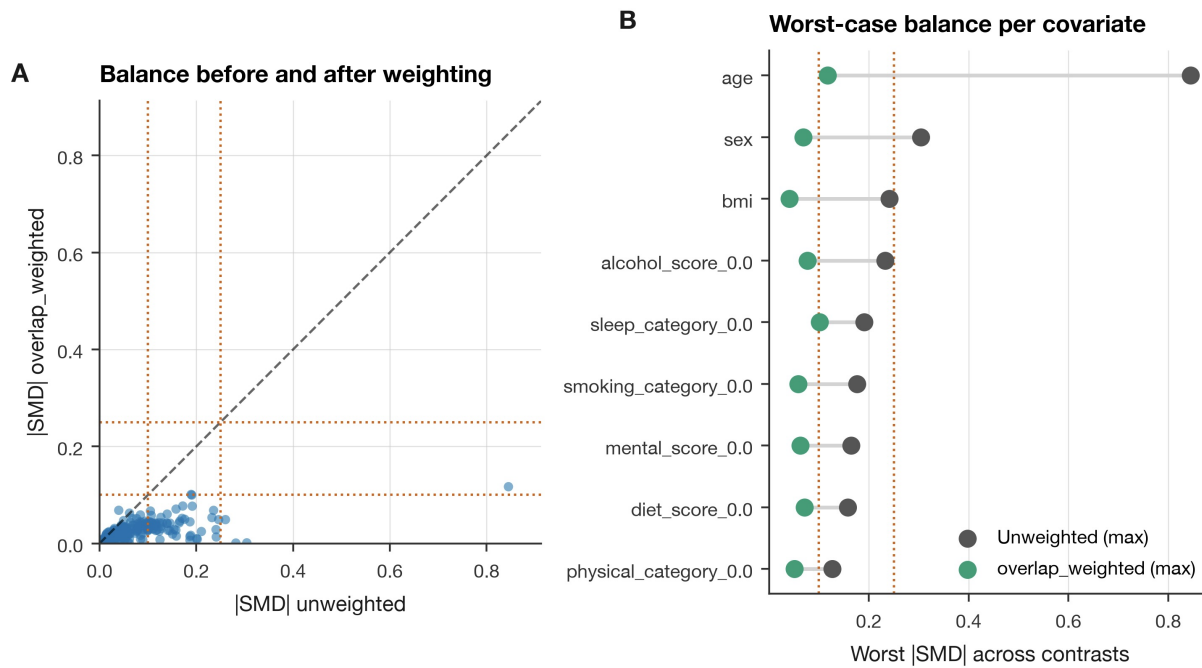


Figure 4.1: Covariate balance before and after overlap weighting: **(A)** covariate-contrast pairs and **(B)** the worst absolute standardised mean difference for each covariate.

Single-variable analysis

Doubly robust average treatment effects were estimated for the 24 pre-specified transitions within the relevant baseline-category strata. One transition was FDR-supported ($q < 0.05$), five were nominally significant only ($p < 0.05$, $q \geq 0.05$), and 18 showed no evidence of an effect (Table 4.2, Figure 4.2). The only FDR-supported finding was the alcohol 2→1 transition, a reduction in consumption from at least weekly to occasional drinking, which was paradoxically associated with a 0.98 pp higher CVD risk (95% CI +0.45 to +1.51, $q = 0.007$). This direction is more consistent with reverse causation, in which deteriorating health prompts reduced drinking, than with a harmful effect of the change itself [37, 38]. Among the nominal findings, mental health was the only domain coherent in both directions, with improvement (1→0) associated with lower risk (−0.95 pp) and deterioration (0→1) with higher risk (+0.70 pp, both $p = 0.014$, $q = 0.114$). The two diet findings were internally inconsistent, with deterioration (2→1, +0.63 pp) and improvement (1→2, +0.52 pp) both associated with higher risk, more consistent with sampling variability than a causal signal, while smoking cessation (2→1, +1.31 pp) was likewise contrary to aetiological expectation, a pattern typical of illness-prompted quitting,

in the domain with the weakest support and overlap.

The remaining 18 transitions showed no evidence of an effect. CI widths separate informative from uninformative nulls: the median CI width was 1.28 pp, and the five
 490 widest intervals all belonged to thin-support transitions. For well-supported transitions, the data therefore exclude effects of observed short-horizon lifestyle change larger than approximately ± 0.5 to 1 pp of absolute risk, whereas the thin-support nulls remain uninformative.

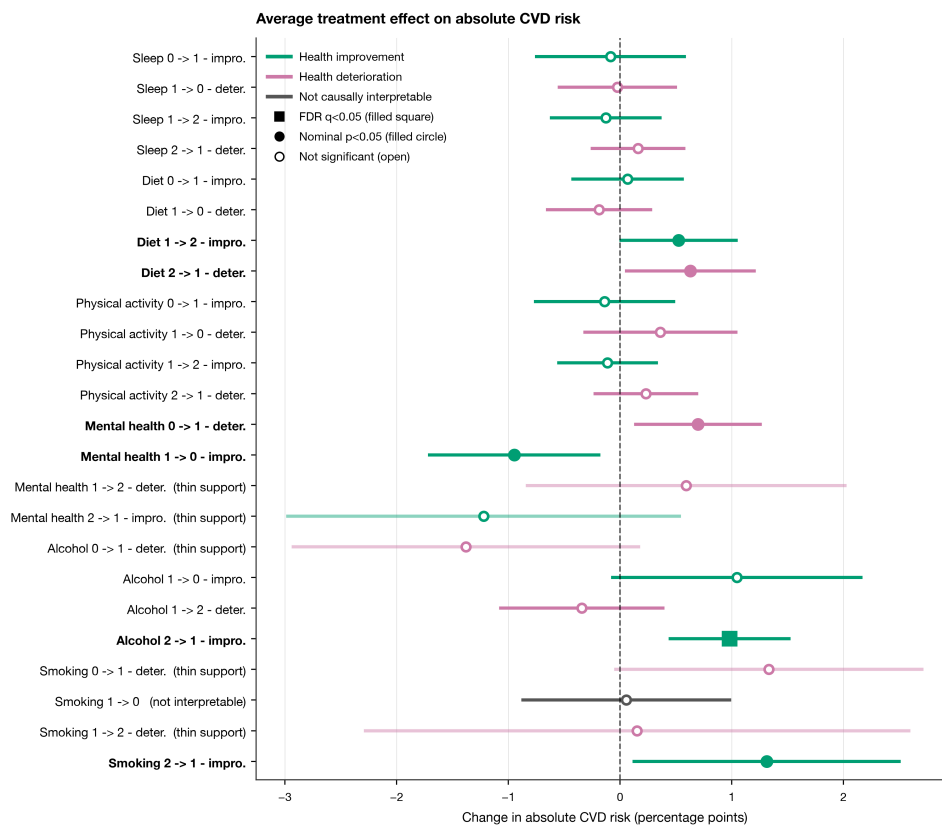


Figure 4.2: Doubly robust average treatment effects and 95% CI for the 24 lifestyle transitions; the filled square denotes FDR support and faded intervals denote thin support.

Effect heterogeneity and subgroup analysis

495 Four complementary checks assessed whether transition effects varied across individuals or subgroups. First, the distributions of conditional average treatment effects were narrow and centred near zero, with a median interquartile range of 0.08 pp and a median signal-to-noise ratio of 0.19, so apparent variation between participants was small relative to estimation noise (Figure S3.3). Second, calibration tests showed FDR-supported

differential calibration for only one transition (alcohol 2→1, $q = 0.002$), while the other differential coefficients were near zero or negative, indicating no usable individual-level signal. Third, best linear projection identified six nominal modifier coefficients among the 96 tested, none surviving FDR correction. Fourth, pre-specified subgroup analyses by sex, age and BMI, tested by Wald tests and participant bootstrap across the 72-test family, yielded seven and six nominal differences respectively, none surviving FDR correction (smallest $q = 0.072$). Taken together, the transition effects were approximately uniform across measured characteristics, and the ATEs summarise the cohort as a whole.

Combined-variable analysis

Multi-arm causal forests compared joint-change arms with participants stable in both domains across the 15 pairwise combinations in improvement, deterioration and mixed scenarios. Because joint arms subdivide the cohort into small cells, the estimability of all 120 contrasts was audited before interpretation: 108 were estimable, and the 12 non-estimable or arithmetically inadmissible contrasts all involved sparse smoking arms (Figure S3.6).

No improvement-scenario contrast survived FDR correction (smallest $q = 0.162$). Nominally protective estimates involved mental-health improvement, alone (-1.61 pp with sleep, -1.96 pp with physical activity) and with both factors improving, consistent with the single-variable mental-health findings but providing no evidence of an additional joint effect. Reduced alcohol consumption within the diet–alcohol model was nominally associated with higher risk ($+1.59$ pp, $p = 0.008$), consistent in direction with the paradoxical single-variable alcohol result. Departures from additivity were small and inconsistent in direction and, as the analysis produced point estimates only, no formal test of interaction was available (Table 4.3, Figure S3.7).

In the mixed scenarios, one contrast within the estimable set survived FDR correction, but it arose from a sparse smoking-related arm: increased alcohol consumption combined with smoking improvement (-4.14 pp, $q = 1.5 \times 10^{-4}$) was based on a single event among 77 participants. A second sparse contrast, sleep deterioration with smoking improvement (-4.30 pp, 3 events among 201), exceeded the reference-arm risk of 4.17%, implying

an impossible negative post-treatment risk, and was excluded as inadmissible by the
530 estimability audit. Both patterns were classified as sparse-cell estimation artefacts rather
than credible joint effects, so no joint-treatment contrast survived FDR correction with
interpretable support (Table 4.3). Companion binary causal forests agreed in direction
for 97 of 108 contrasts, so the conclusions did not depend on the multi-arm specification,
although the two designs share participants.

535 Sensitivity analyses

All models were re-estimated using only events diagnosed after the imaging visit. This
restriction reduced the outcome to 129 events among 69,105 participants (0.19%), so every
transition is effectively thin support. The FDR-supported alcohol 2→1 finding did not
persist (-0.02 pp, 95% CI -0.13 to $+0.09$), indicating that events occurring before the
540 second assessment contributed materially to the primary estimate, exactly the pattern
expected if declining health both prompted reduced drinking and preceded diagnosis.
Other transitions showed nominal intervals excluding zero, but at this event count the
estimates are unstable and were not multiplicity-corrected (Figure S3.8).

Every contrast was re-estimated under five estimands, comprising the full-sample primary,
545 overlap weighting and propensity trimming at three thresholds. The sign was stable
across all five estimands for 21 of 24 contrasts, and the three exceptions are individually
dismissible. Trimming-induced sample collapse was confined to the smoking domain and is
reported as a positivity diagnostic. The FDR-supported alcohol finding moved from $+0.98$
pp to $+0.90$ pp (overlap-weighted) and $+0.67$ pp (strictest trimming), so poor overlap
550 does not explain it (Figure S3.9).

E-values were computed on the risk-ratio scale, the minimum association a confounder
would need with both the transition and CVD, conditional on measured covariates, to
explain away an estimate. Arm-specific risks came from the doubly robust AIPW scores
with 1,000-replicate bootstrap intervals, and E-values were estimable for all 24 transitions.
555 For the alcohol 2→1 finding, the implied risk ratio was 1.30 (95% CI 1.14 to 1.47), giving
an E-value of 1.92 at the point estimate and 1.54 at the confidence limit (Figure S3.10).

To examine the estimates under specified patterns of residual confounding, seven pre-

specified bias scenarios varied the confounder's prevalence imbalance between arms and its risk ratio with CVD. The alcohol 2→1 estimate was unchanged under the null-control scenario, remained positive and significant under weak imbalance (+0.89 pp), attenuated with an interval including zero under moderate imbalance (+0.50 pp), and reversed only under the strong scenario (Figure 4.3). Tipping-point analysis gave a similar threshold: a risk-increasing confounder concentrated among movers, such as deteriorating baseline health, would require a risk ratio of at least 2.6 to null the estimate.

Taken together, the sensitivity analyses exclude poor overlap as an explanation, while the disappearance of the finding under the strict temporal design and its vulnerability to moderate confounding by declining health support a reverse-causation reading rather than a causal effect of reduced drinking.

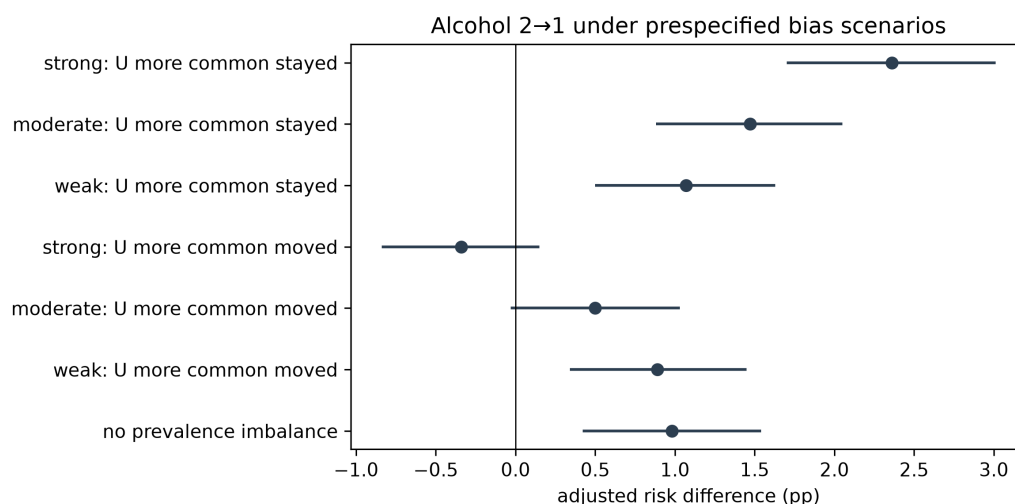


Figure 4.3: Adjusted risk difference for the alcohol 2 → 1 transition under pre-specified unmeasured-confounding scenarios.

External validation

No contrast reached nominal or FDR-adjusted significance (smallest $p = 0.077$). Effect directions agreed with UKB for 8 of 11 contrasts, and all corresponding confidence intervals overlapped, although NHEFS estimates were imprecise. The paradoxical alcohol direction recurred: the 2 → 1 transition was +0.56 pp in NHEFS versus +0.98 in UKB, while 1 → 0 was +2.41 versus +1.04. Two of the three disagreements involved smoking; cessation was protective in NHEFS (−0.52 pp), contrasting with the anomalous UKB estimate (+1.31)

(Table S3.2). Overall, NHEFS provided limited directional consistency, particularly for alcohol, but insufficient precision to confirm effect-size transportability.

4.3 Visualisation tool

The predictive and causal results were integrated into an interactive Streamlit research
580 prototype (Figure 4.4). Users enter demographic and lifestyle information, after which the
tool displays estimated 10-year CVD risk, compares it with the cohort average, and presents
single- and paired-lifestyle effect estimates with 95% CI. A summary plan distinguishes
changes for which the study found no clear evidence of benefit from behaviours already
in the healthier category. Explanatory panels clarify the distinction between prediction
585 and causation and state that the outputs are observational research estimates rather than
clinical recommendations.

Table 4.3: Selected combined-variable treatment-effect estimates.

Pair	Scenario	Arm	n	Events	ATE (pp)	95% CI (pp)	p	q (FDR)
Alcohol $\times$ smoking	Mixed	Alcohol deteriorates + smoking improves	77	1	-4.14	[-5.81, -2.46]	1.3×10^{-6}	1.5×10^{-4}
PA $\times$ alcohol	Mixed	PA improves + alcohol deteriorates	436	6	-1.93	[-3.21, -0.64]	0.003	0.124
Diet $\times$ alcohol	Improvement	Alcohol improves only	3,592	157	+1.59	[+0.42, +2.76]	0.008	0.162
PA $\times$ mental	Improvement	Mental improves only	2,484	80	-1.96	[-3.43, -0.48]	0.009	0.162
Sleep $\times$ mental	Improvement	Sleep + mental both improve	2,087	77	-1.74	[-3.06, -0.42]	0.010	0.162
Sleep $\times$ mental	Improvement	Mental improves only	3,345	120	-1.61	[-2.84, -0.38]	0.010	0.162
Alcohol $\times$ smoking	Improvement	Alcohol improves only	2,323	119	+1.56	[+0.28, +2.83]	0.017	0.238
PA $\times$ mental	Improvement	PA + mental both improve	2,818	104	-1.60	[-3.05, -0.14]	0.032	0.330
PA $\times$ alcohol	Improvement	Alcohol improves only	2,285	95	+1.25	[+0.09, +2.41]	0.034	0.330
Mental $\times$ alcohol	Improvement	Mental improves only	6,520	225	-0.87	[-1.69, -0.05]	0.037	0.330
Mental $\times$ smoking	Deterioration	Mental deteriorates only	6,187	219	+0.67	[+0.04, +1.30]	0.038	0.330

Note: ATEs and CI are expressed as percentage-point differences in absolute CVD risk. FDR-adjusted q -values were calculated using the Benjamini–Hochberg procedure.

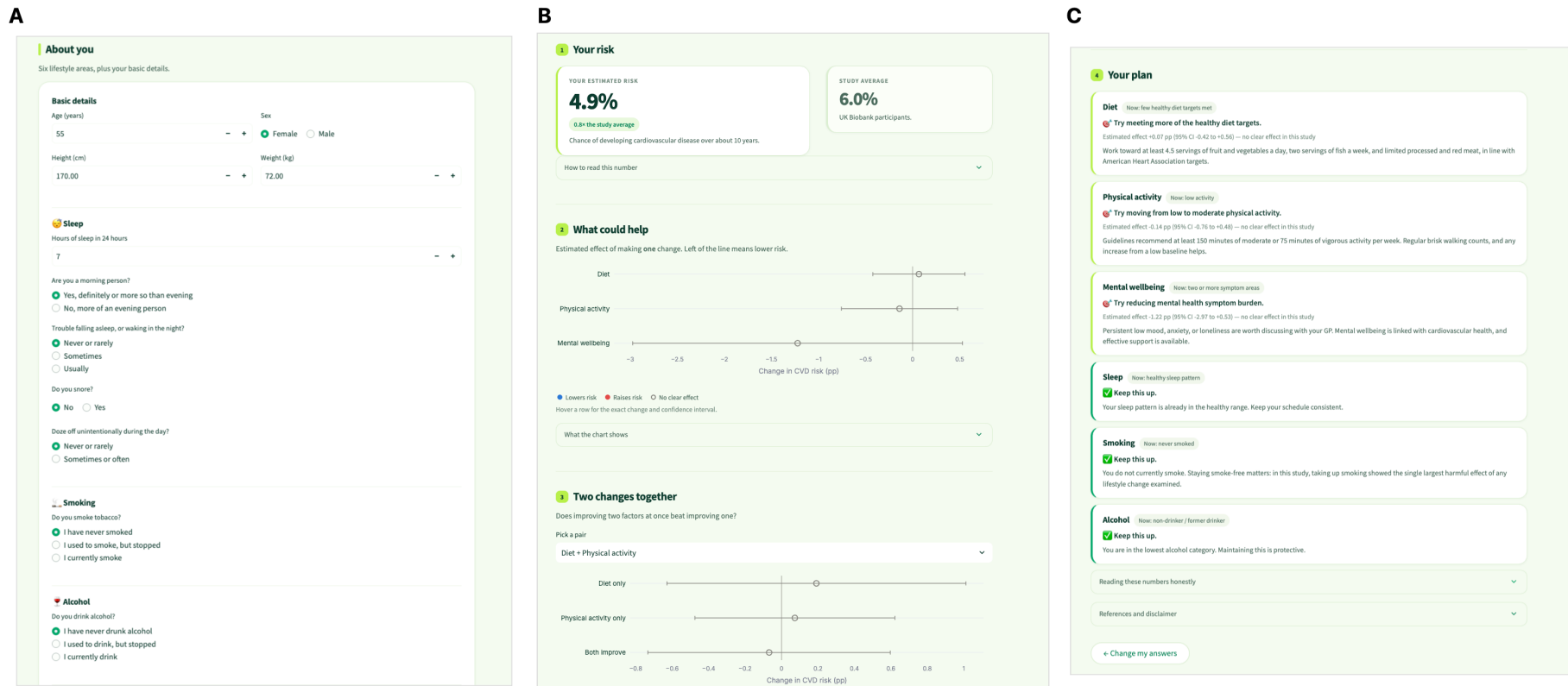


Figure 4.4: Streamlit research prototype showing (A) demographic and lifestyle inputs, (B) predicted CVD risk and estimated lifestyle effects, and (C) the summary plan for an illustrative profile.

5. DISCUSSION

This study developed complementary predictive and causal machine-learning models of lifestyle and cardiovascular risk using UK Biobank data, combining baseline risk prediction with a timing-aware causal analysis of observed lifestyle change and external validation in an independent cohort. Four prediction models achieved similar discrimination, calibration and clinical utility, so logistic regression was selected for interpretability. Across 24 single-variable transitions and 108 estimable joint contrasts, most showed no evidence of an effect, and for transitions with adequate support the confidence intervals were narrow enough to rule out risk changes larger than approximately one pp. Only one finding survived correction for multiple testing, namely that participants who reduced their alcohol consumption from at least weekly to occasional drinking had a 0.98 pp higher subsequent CVD risk, a direction opposite to the expected benefit of drinking less. Mental health was the only domain in which improvement and deterioration were associated with lower and higher risk respectively in both analyses. As examined below, the sensitivity analyses indicate that the paradoxical alcohol association reflects reverse causation rather than a causal effect of the change.

The AUC-ROC of 0.718 (95% CI 0.712 to 0.724) from age, sex, BMI and six lifestyle variables is below clinically enriched calculators such as QRISK3 (0.851) and SCORE2 (0.742), which include blood pressure, lipids and comorbidities [39, 40], and is consistent with models based mainly on lifestyle information [41, 42]. QRISK3 and SCORE2 require blood pressure, lipids and clinical history, whereas the present model relies entirely on self-reported lifestyle and basic demographic information, so good discrimination from freely obtainable inputs is itself practically useful where clinical testing is unavailable. The largely null effects of lifestyle change are also unsurprising, because the cardiovascular benefits of changes such as smoking cessation take 5 to 15 years to appear, longer than the interval between T0 and T2 [43]. The mental-health results fit meta-analytic evidence linking depression and anxiety to higher risk of incident CVD [44]. Previous studies typically predicted risk from a few lifestyle variables measured once, rather than modelling comprehensive lifestyle information and its change [42]. The present study combines both in one framework and explicitly accounts for whether CVD events occurred before or after the imaging visit.

A baseline-level reference analysis provided additional context for the transition estimates. Healthier baseline levels of sleep, diet, smoking, physical activity and mental health were associated with lower CVD risk, whereas lower baseline alcohol consumption was associated with higher risk, reproducing the paradoxical direction of the longitudinal alcohol transition. Because baseline lifestyle level and lifestyle change represent different exposures, this comparison provides only qualitative directional evidence. Nevertheless, the persistence of the alcohol paradox at baseline suggests that reverse causation arising exclusively during the T0–T2 interval is unlikely to be its sole explanation.

The strongest signal, higher CVD risk among participants who reduced their drinking, survived correction for multiple testing and remained stable under overlap weighting and trimming. However, evidence concerning alcohol and cardiovascular risk is heterogeneous. Some observational studies report that moderate alcohol consumption is associated with lower risk at particular consumption levels [45], whereas other studies suggest that these apparent benefits may reflect differences in baseline health, lifestyle and socioeconomic characteristics between drinking groups [46, 47]. Mendelian-randomisation studies have instead generally indicated that greater alcohol consumption increases cardiovascular risk [48]. Some studies also suggest that low-moderate alcohol use might be harmful too [49].

The alcohol classification used here cannot resolve this uncertainty. The lowest category combined never and former drinkers, the highest combined weekly and daily drinking, and frequency did not capture quantity, beverage type or binge drinking, so the 2→1 contrast pooled several distinct reductions in alcohol exposure and cannot identify a specific frequency as beneficial or harmful. The association disappeared in the strict post-imaging analysis, but with only 129 events this analysis cannot distinguish temporal attenuation from loss of power. Its E-value of 1.92 and attenuation under the bias scenarios also indicate that residual confounding could plausibly explain the result. The finding is therefore compatible with reverse causation, baseline confounding, exposure misclassification or a non-linear association. The present data cannot distinguish these explanations and provide no evidence that reducing alcohol consumption is harmful. Mental health showed the next clearest pattern, as the only domain in which improvement was associated with lower risk (−0.95 pp) and deterioration with higher risk (+0.70 pp), a coherent pattern that reappeared in the combined analyses.

The remaining transitions showed no evidence of an effect, yet these nulls are informative rather than disappointing. For well-supported transitions the confidence intervals rule out effects larger than about ± 0.5 to 1 pp, rather than merely failing to detect them. Two-point measurement also dilutes gradual change through regression dilution, and a one-step category change is a modest exposure. The counter-expected smoking-cessation signal (+1.31 pp) is best explained by the same reverse-causation mechanism, since illness commonly prompts quitting, and it lost all evidence of an effect under the strict design.

Because treatment effects were approximately uniform over measured characteristics, the interactive tool applies average rather than personalised estimates by design. Lower discrimination in participants aged 60 and over reflects the restricted ranking information age carries in an older stratum and accumulating unmeasured clinical risk.

The primary strength is the timing-aware design, which fitted three temporal models, namely the broad primary window, the strict post-imaging window and the baseline reference, so reverse causation was anticipated by design. Estimands were pre-specified under target-trial emulation with doubly robust estimation and honest forests, estimability and support were audited openly, and sensitivity coverage spanned temporality, overlap and unmeasured confounding. A further strength is that the causal estimates were externally validated in an independent US cohort spanning a different population and historical period. The paradoxical alcohol direction recurred in that cohort, which suggests that illness-prompted reductions in drinking are a general feature of observational lifestyle-change data rather than an artefact specific to UK Biobank.

Despite these strengths, the design is observational, and the adjustment set lacks medication history, prevalent disease, socioeconomic position and family history, so unmeasured confounding cannot be excluded. The E-values, bias scenarios and baseline reference model bound this threat rather than remove it. Lifestyle was measured only twice, so changes cannot be dated and short-lived changes may be missed, and the broad primary window counts events occurring before the change was complete. The strict post-imaging analysis addresses this but leaves too few events to be conclusive. UK Biobank participants are healthier than the general population and the imaging subsample is healthier still, so the causal cohort is a selected group and generalisability is limited. Finally, the analysis relied on demographic and lifestyle variables alone, whereas

CVD risk is also strongly shaped by genetic predisposition, so the measured covariates capture only part of what drives both lifestyle change and CVD.

Future work follows from these limitations. First, longer follow-up with repeated lifestyle assessments would date changes and provide enough post-measurement events to make the strict design conclusive. Second, future alcohol analyses should distinguish never from former drinkers and weekly from daily consumption, incorporate quantity and beverage type, and examine cardiovascular outcomes separately [45]. Third, future models should incorporate genetic data to capture inherited CVD risk and examine gene-lifestyle interplay. Fourth, richer clinical and socioeconomic covariates would narrow residual confounding. Finally, validation should be applied in more representative populations to test generalisability beyond UK Biobank.

In this large prospective cohort, lifestyle information supported good discrimination of cardiovascular risk, while observed short-horizon lifestyle changes showed little evidence of altering that risk. The single association that survived multiple-testing correction was identified, through temporal and confounding scrutiny, as reverse causation rather than a treatment effect, which demonstrates the value of building timing awareness into the design rather than discovering it afterwards. By integrating prediction, timing-aware causal inference, external validation and interactive risk communication within one framework, this study treats prediction and causation as complementary rather than interchangeable questions, and provides a template that longer follow-up with repeated lifestyle assessment can extend directly.

6. ETHICAL CONSIDERATIONS

6.1 Data

The North West Multi-centre Research Ethics Committee granted ethical approval for the UKB, and all participants provided informed consent, including consent for linkage
705 to health records. This research was conducted using the UK Biobank Resource under Application Number 12113. Data were de-identified at source, analysed on approved secure systems with no attempt at re-identification, and are reported only in aggregate. No participants were contacted and no new data were collected. All analyses comply with UKB's data access and publication policies, exempting this study from additional ethical
710 approval.

6.2 Web Application

The tool is a research prototype, not a diagnostic instrument, and a clear disclaimer states that its suggestions do not substitute clinical advice. Because it reports average differences among participants with similar measured characteristics rather than guaranteed individual
715 benefit, every estimate is displayed with its confidence interval so that users are not misled by apparent precision. Its accessibility primarily to younger, digitally proficient adults risks widening health inequalities among older and socioeconomically disadvantaged populations, a concern compounded in older users for whom model discrimination was lower; this will be addressed through inclusive interface design such as an audio version. Wording
720 throughout presents lifestyle as one modifiable contributor rather than as individual responsibility.

7. AI DECLARATION

Generative AI tools were used only to support spelling and grammar review and the development of the Streamlit web application. They were not used to select the research
725 question, conduct the statistical analyses, interpret the results, or construct the predictive or causal models. All AI-assisted material was reviewed and verified by the author, who takes full responsibility for the submitted content, code, analyses and conclusions.

SUPPLEMENTARY MATERIAL

S1. SUPPLEMENTARY METHODS

Table S1.1: Characteristics of the baseline and imaging cohorts

Characteristic	Baseline cohort (<i>n</i> = 458,840)	Baseline + imaging visit (<i>n</i> = 71,428)
<i>N</i>	458,840	71,428
Age, years, mean (SD)	56.2 (8.1)	54.6 (7.5)
Female, <i>n</i> (%)	249,956 (56.1)	36,605 (51.2)
BMI, kg/m ² , mean (SD)	27.2 (4.6)	26.5 (4.2)
Sleep category 0, <i>n</i> (%)	100,311 (21.9)	13,985 (19.6)
Sleep category 1, <i>n</i> (%)	205,826 (44.9)	31,364 (43.9)
Sleep category 2, <i>n</i> (%)	151,798 (33.1)	26,079 (36.5)
Diet category 0, <i>n</i> (%)	156,261 (34.1)	23,983 (33.6)
Diet category 1, <i>n</i> (%)	216,635 (47.3)	34,342 (48.1)
Diet category 2, <i>n</i> (%)	85,035 (18.6)	13,103 (18.3)
Physical activity category 0, <i>n</i> (%)	108,398 (23.8)	15,819 (22.1)
Physical activity category 1, <i>n</i> (%)	180,362 (39.6)	29,161 (40.8)
Physical activity category 2, <i>n</i> (%)	167,223 (36.7)	26,448 (37.0)
Mental health category 0, <i>n</i> (%)	344,651 (75.3)	57,122 (80.0)
Mental health category 1, <i>n</i> (%)	82,953 (18.1)	11,067 (15.5)
Mental health category 2, <i>n</i> (%)	29,877 (6.5)	3,239 (4.5)
Alcohol category 0, <i>n</i> (%)	34,551 (7.6)	3,380 (4.7)
Alcohol category 1, <i>n</i> (%)	102,305 (22.4)	13,420 (18.8)
Alcohol category 2, <i>n</i> (%)	320,702 (70.1)	54,628 (76.5)
Smoking category 0, <i>n</i> (%)	255,429 (56.0)	43,493 (60.9)
Smoking category 1, <i>n</i> (%)	153,380 (33.6)	23,253 (32.6)
Smoking category 2, <i>n</i> (%)	47,549 (10.4)	4,682 (6.6)
Incident CVD events, <i>n</i> (%)	27,454 (5.98)	2,452 (3.43)

S1.1 Detailed lifestyle variable definitions

Lifestyle variables were derived identically at the baseline assessment (T0) and imaging assessment (T2). UK Biobank response codes −1 (“do not know”) and −3 (“prefer not to answer”) were generally treated as missing. Domain-specific exceptions and partial-data rules are described below. The derived domain tables were merged with demographic and outcome data by participant identifier (`eid`).

Sleep [19]. The healthy sleep score used five UK Biobank touchscreen fields: sleep duration (Field 1160), chronotype (1180), sleeplessness/insomnia (1200), snoring (1210), and daytime dozing (1220). One point was assigned for each available component meeting its healthier criterion: sleeping 7–8 hours per day; reporting a definite or relative morning
 740 chronotype; reporting sleeplessness/insomnia never or rarely; not snoring; and reporting daytime dozing never/rarely or sometimes. The component score was divided by the number of non-missing components and rescaled to a range of 0–5:

$$\text{sleep score} = 5 \times \frac{\text{number of healthier components}}{\text{number of observed components}}.$$

The score was missing only when all five components were unavailable. Rescaled scores ≤ 2 were classified as category 0, scores greater than 2 but less than 4 as category 1, and
 745 scores ≥ 4 as category 2. Higher categories therefore represented healthier sleep.

Diet [21, 22, 23]. Diet was constructed from cooked-vegetable intake (Field 1289), salad/raw- vegetable intake (1299), fresh-fruit intake (1309), oily-fish intake (1329), non-oily-fish intake (1339), processed-meat intake (1349), and beef, lamb/ mutton, and pork intake (1369, 1379, and 1389). Three equally weighted criteria contributed one point each:

- 750 1. fresh-fruit intake plus one third of the summed cooked- and raw-vegetable intake was at least 4.5 questionnaire-scale units.
2. summed oily- and non-oily-fish intake was at least 2.
3. processed-meat intake was no more than 2 and summed beef, lamb/mutton, and pork intake was no more than 5.

755 The resulting score ranged from 0 to 3 and was missing only when all dietary components were unavailable. Otherwise, a criterion with insufficient information contributed no point. For the three-level analysis, original scores 0–1 were mapped to category 0, score 2 to category 1, and score 3 to category 2. Higher categories represented greater adherence to the dietary criteria.

760 **Alcohol** [24]. Alcohol was derived by combining alcohol-drinker status (Field 20117) with reported intake frequency (Field 1558). An initial five-level variable was constructed: 0

for never drinkers, 1 for former drinkers, 2 for current drinkers reporting consumption on special occasions or one to three times per month, 3 for current drinkers reporting consumption once or twice or three to four times per week, and 4 for current drinkers reporting daily or almost-daily consumption. Participants whose frequency response indicated never drinking but whose status was coded as “prefer not to answer” were classified as never drinkers; other missing or incompatible combinations remained missing.

For consistency with the other domains, this variable was subsequently collapsed to three levels: never and former drinkers (original levels 0–1) formed category 0; special-occasion or monthly drinkers (level 2) formed category 1; and weekly or daily drinkers (levels 3–4) formed category 2. Higher values therefore represented greater reported drinking frequency, rather than a directly measured alcohol dose or an unambiguous ordering of cardiovascular health. The score did not distinguish beverage type, quantity per occasion, or binge drinking.

Smoking [25]. Smoking was obtained directly from the derived UK Biobank smoking-status variable (Field 20116), which combines current and previous tobacco-smoking responses. Participants were classified as never smokers (category 0), former smokers (category 1), or current smokers (category 2). “Prefer not to answer” was treated as missing. Higher categories represented increasing smoking burden.

Physical activity [26, 27, 28]. Physical activity was calculated from the reported number of days per week and minutes per day spent walking (Fields 864 and 874), undertaking moderate-intensity activity (884 and 894), and undertaking vigorous-intensity activity (904 and 914). “Do not know” and “prefer not to answer” were treated as missing, while responses indicating less than 10 minutes were set to zero. When activity days were reported but duration was missing, a duration of 10 minutes was assigned.

Weekly metabolic-equivalent minutes were calculated as

$$\begin{aligned}\text{walking MET-min/week} &= 3.3 \times \text{minutes/day} \times \text{days/week}, \\ \text{moderate MET-min/week} &= 4.0 \times \text{minutes/day} \times \text{days/week}, \\ \text{vigorous MET-min/week} &= 8.0 \times \text{minutes/day} \times \text{days/week}.\end{aligned}$$

The three values were summed to obtain total MET-minutes per week.

Participants were assigned to the high-activity category (2) if they reported either at least three days of vigorous activity and at least 1,500 MET-minutes per week, or at least 3,000 MET-minutes across seven or more summed activity-days. The moderate category (1) required at least one of the following: vigorous activity for at least 20 minutes on three days; moderate activity for at least 30 minutes on five days; walking for at least 30 minutes on five days; or at least 600 MET-minutes across five or more summed activity-days. Participants with usable responses who met neither criterion were assigned to the low category (0). Higher categories represented greater physical activity.

Mental health [19]. Mental health was constructed from frequency of depressed mood (Field 2050), unenthusiasm/disinterest (2060), tenseness/restlessness (2070), tiredness/lethargy (2080), and loneliness/isolation (2020). Responses to the four symptom-frequency items were recoded from 1–4 to 0–3. Depressed mood and unenthusiasm/disinterest were summed, with one point assigned when their combined score was at least 3. Tenseness/restlessness and tiredness/lethargy were similarly summed, with one point assigned at a combined score of at least 3. A further point was assigned when loneliness was reported.

The resulting symptom-burden score ranged from 0 to 3 and was missing only when all three components were unavailable. For the three-level analysis, scores 0 and 1 were retained as categories 0 and 1, while scores 2 and 3 were combined as category 2. Higher categories represented greater mental- health symptom burden.

S1.2 Predictive model tuning and evaluation settings

For logistic regression, L2 regularisation strength was selected by grid search over $C \in \{0.01, 0.1, 1, 10, 100\}$ with the lbfgs optimiser and five-fold stratified cross-validation. Tree-based models used randomised search over eight configurations sampled from each algorithm’s predefined space with three-fold cross-validation. This balanced breadth of model-complexity evaluation against computational feasibility. Tree-based models received no scaling or spline transformation, as these algorithms are invariant to monotonic feature transformations. Calibration slope and intercept were estimated by regressing observed outcomes on the logit of predicted probabilities, with ideal values of 1.0 and 0.0 respectively.

Table S1.2: Hyperparameter tuning for the four prediction models. All searches optimised AUC–ROC within the development set.

Model	Search and CV	Hyperparameter space
Logistic regression	Grid search; 5-fold stratified	L2 penalty; $C \in \{0.01, 0.1, 1, 10, 100\}$; solver = <code>lbfgs</code> .
Random forest	Randomised search ($n = 8$); 3-fold	<code>n_estimators</code> $\in \{300, 500, 800\}$; <code>min_samples_leaf</code> $\in \{20, 50, 100\}$; <code>max_features</code> $\in \{\sqrt{p}, 0.5\}$.
XGBoost	Randomised search ($n = 8$); 3-fold	<code>n_estimators</code> $\in \{400, 600, 800\}$; learning rate $\in \{0.03, 0.05, 0.1\}$; depth $\in \{3, 4, 6\}$; subsample and column sample $\in \{0.8, 1.0\}$.
LightGBM	Randomised search ($n = 8$); 3-fold	<code>n_estimators</code> $\in \{400, 600, 800\}$; learning rate $\in \{0.03, 0.05, 0.1\}$; leaves $\in \{31, 63, 127\}$; subsample and column sample $\in \{0.8, 1.0\}$.

S1.3 Baseline causal reference model

The transition analyses estimate the effect of changing lifestyle between T0 and T2 and are therefore exposed to reverse causation. A participant who develops symptoms before the T2 assessment may change behaviour in response, indicating a timing bias that no covariate adjustment can remove. A separate and temporally clean reference model was therefore fitted in which the exposure is baseline lifestyle level (T0) rather than change. For each of the six domains, treatment was 1 if the participant's baseline level fell in the healthy category and 0 otherwise. A binary causal forest was fitted per domain with the same adjustment set, honest estimation, and local centering as the transition analyses. Because this model requires no repeat assessment at T2, it uses the full cohort ($N = 458,316$; 27,384 incident events, 6.0%). Because this model requires no repeat assessment at T2, it uses the baseline cohort after one further exclusion. A total of 524 participants free of CVD under the primary definition but with atrial fibrillation or heart failure recorded before baseline were removed, so that the reference model estimates effects in a cohort free of prevalent cardiovascular disease under the broader composite definition, leaving $N = 458,316$ with 27,384 incident events (6.0%). It cannot answer the question the transition analyses address, but it provides an independent check on whether their effect directions are credible. Agreement supports the transition results, while disagreement, or agreement on a paradoxical direction, localises the likely source of bias.

S1.4 Honest estimation settings

All forests were fitted with honesty enabled and grf’s default honesty fraction of 0.5, in which half the sample determines tree structure (which variables to split on, and where) and the disjoint half estimates the within-leaf treatment effects. A CI group size of 2 was
 840 used for confidence interval construction.

S1.5 Calibration of CATE predictions

Individuals were ranked by predicted CATE and divided into quartiles (Q1 lowest to Q4 highest predicted benefit). Within each quartile the ATE was re-estimated with the doubly robust estimator, the quartile ATEs were plotted against the mean predicted CATEs,
 845 and a linear regression of quartile ATE on predicted CATE was fitted, reporting the calibration slope and R^2 . A slope near 1 with high R^2 indicates well-calibrated predictions. A positive slope with moderate R^2 indicates genuine but noisy heterogeneity. A slope near 0 indicates the predicted heterogeneity is uninformative. Because the quartiles are defined by the same forest that produces the predictions, this is an exploratory adaptive analysis
 850 rather than an independent validation, and it is labelled as such wherever reported. As a complementary check, the RATE test [50] compared doubly robust ATEs in the top and bottom 20% of the predicted-CATE distribution. This output was subsequently excluded from the results for the inadmissibility reason described in the Methods.

S1.6 Seed stability

855 The primary analysis was run with a single random seed (42), so the seed-stability output records the primary fit only and does not demonstrate stability across random initialisations. This is listed as a limitation. The recorded per-transition standard deviations of the CATE distributions are nonetheless informative, being very small (e.g. 5.3×10^{-4} for the sleep 0→1 transition) and confirming the narrow CATE distributions
 860 reported in the Results.

S1.7 Unmeasured confounding implementation (E-values and QBA)

E-values [51] were computed from the closed form for both the point estimate and the confidence limit closest to the null, converting risk differences to approximate risk ratios via the model-based reference risk. The QBA modelled a generic binary unmeasured confounder U . Each scenario specifies the residual prevalence of U in the mover and stayer arms together with a common risk-ratio association of U with CVD. Rather than subtracting a bias term, both arm risks are standardised to a common prevalence of U (each observed arm risk is first stripped of U 's contribution and then recombined under the arm-averaged prevalence) and the adjusted risk difference is recomputed. Seven pre-specified scenarios covered both directions of prevalence imbalance plus a no-imbalance null control. The analysis assumes a binary U , a common risk ratio in both arms, no U -by-treatment interaction, and no other bias mechanism operating simultaneously. It produces point estimates only, without adjusted confidence intervals. Both procedures were applied to the primary cohort and therefore accompany the primary ATEs, complementing the strict-temporal analysis that addresses outcome timing separately.

S1.8 NHEFS harmonisation caveats

BMI in NHEFS was approximated from measured 1982 weight and sex-specific mean height, as NHEFS records no adult height. Physical activity was measured with different instruments in 1982 and 1987 and was collapsed to a coarse three-level scale, so some apparent transitions may reflect instrument change rather than behaviour change. Physical-activity results are interpreted with this caveat.

S2. SUPPLEMENTARY RESULTS — PREDICTIVE MODELLING

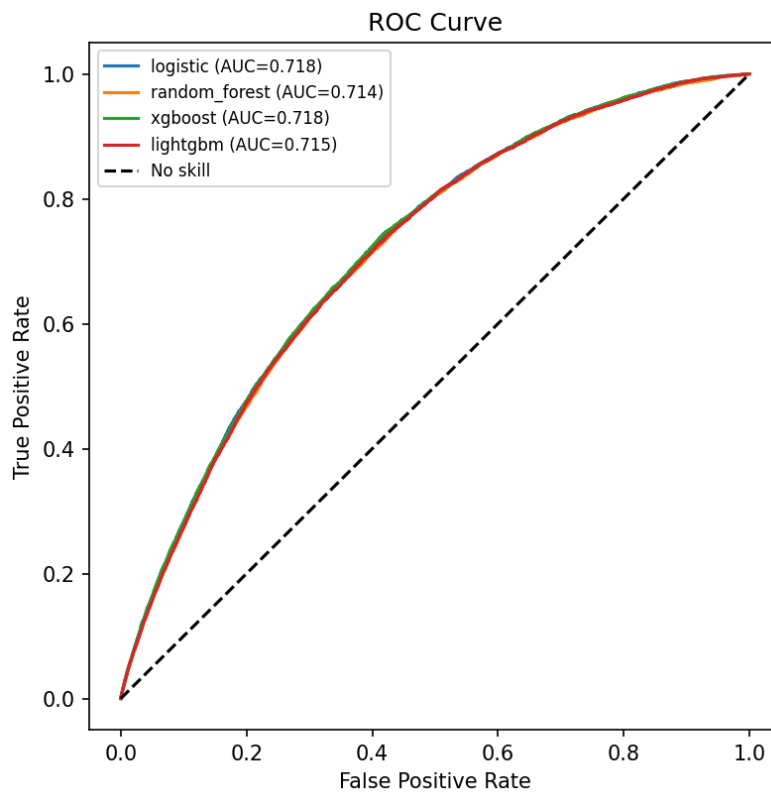


Figure S2.1: Receiver operating characteristic curves for the four CVD prediction models; the dashed line represents a no-skill classifier.

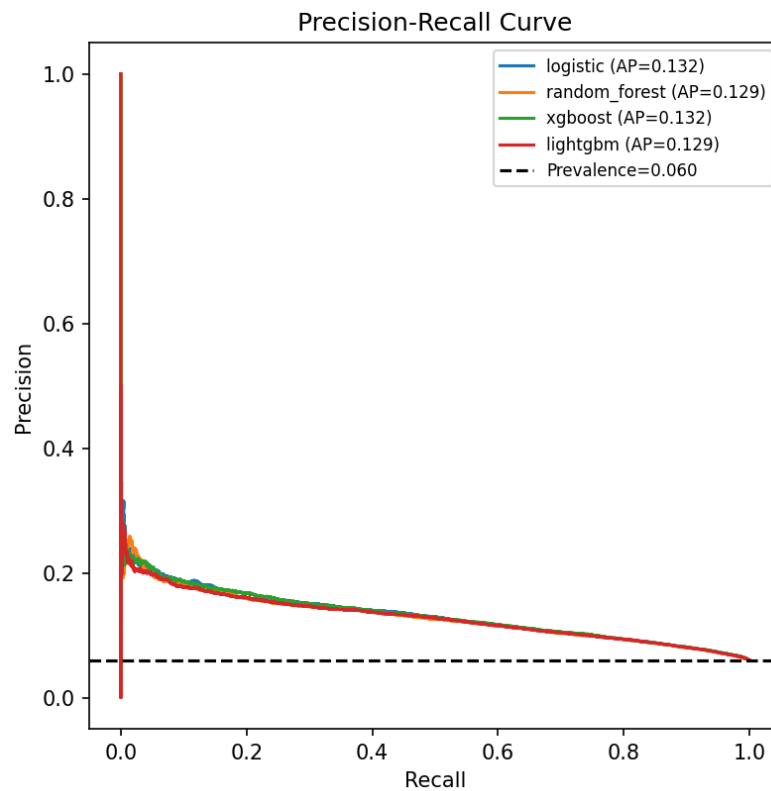


Figure S2.2: Precision–recall curves for the four CVD prediction models; the dashed line indicates the event prevalence.

Table S2.1: Predictive performance and DeLong comparisons in the secondary outcome analysis.

Model	AUC (95% CI)	AP	Brier	Calibration slope/intercept	Δ AUC vs LR (DeLong p)
Logistic regression	0.7155 (0.7104–0.7205)	0.1747	0.0715	0.963/–0.0056	Reference
Random forest	0.7132 (0.7081–0.7183)	0.1738	0.0715	0.981/–0.0052	–0.0023 ($p = 3.88 \times 10^{-5}$)
XGBoost	0.7165 (0.7115–0.7213)	0.1755	0.0714	0.954/–0.0054	+0.0010 ($p = 0.015$)
LightGBM	0.7143 (0.7092–0.7191)	0.1729	0.0715	0.923/–0.0045	–0.0012 ($p = 0.033$)

Note: AP, average precision; LR, logistic regression. Calibration values are presented as slope/intercept.

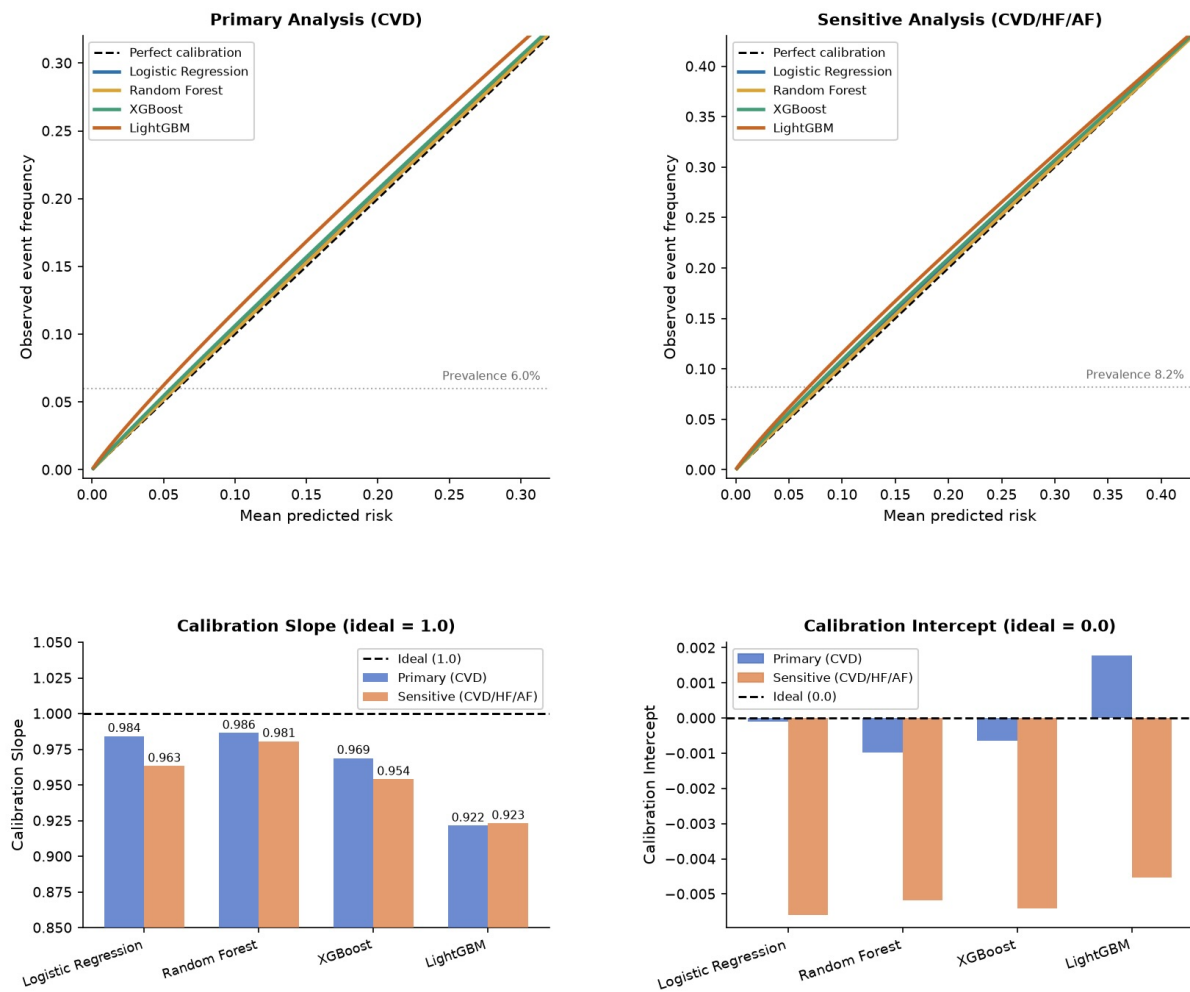


Figure S2.3: Calibration of the four prediction models in the primary CVD and secondary outcome CVD/HF/AF analyses. Top panels show observed versus predicted risk; bottom panels show calibration slopes and intercepts. Dashed lines indicate perfect calibration and dotted lines indicate outcome prevalence.

Table S2.2: Tree-based feature importance in the primary and secondary outcome analyses.

Feature	Random forest		XGBoost		LightGBM	
	Primary	Secon.	Primary	Secon.	Primary	Secon.
Age	0.410	0.484	0.266	0.331	5,600	5,054
BMI	0.208	0.198	0.045	0.048	6,777	7,495
Alcohol score	0.044	0.030	0.041	0.029	2,383	2,209
Diet score	0.028	0.022	0.020	0.017	1,676	1,721
Mental health score	0.038	0.028	0.064	0.061	1,850	1,854
Sleep category	0.029	0.024	0.044	0.035	1,534	1,449
Smoking category	0.085	0.063	0.149	0.119	1,762	1,720
Physical activity	0.028	0.023	0.023	0.019	1,468	1,501
Sex	0.130	0.127	0.349	0.341	950	997

Note: Secon., secondary outcome analysis. Random-forest and XGBoost values are relative importance scores; LightGBM values are split counts. Values should not be compared directly across algorithms.

Table S2.3: Logistic-regression coefficients and odds ratios in the primary analysis.

Predictor	Coefficient	Odds ratio
Age spline basis 0	−9.889	5.08×10^{-5}
BMI spline basis 0	3.446	31.359
Age spline basis 1	−2.731	0.065
Age spline basis 2	−1.800	0.165
BMI spline basis 1	−1.603	0.201
Age spline basis 3	−0.804	0.447
Smoking category 2 vs 0	0.711	2.035
Sex	0.705	2.024
BMI spline basis 2	−0.416	0.660
BMI spline basis 4	0.373	1.452
Age spline basis 4	0.235	1.265
Sleep category 2 vs 0	−0.199	0.820
Smoking category 1 vs 0	0.193	1.213
BMI spline basis 3	−0.173	0.841
Mental-health score	0.141	1.152
Alcohol score	−0.118	0.889
Physical-activity category 2 vs 0	−0.110	0.896
Physical-activity category 1 vs 0	−0.102	0.903
Sleep category 1 vs 0	−0.075	0.928
Diet score	−0.047	0.954

Note: Odds ratios equal $\exp(\text{coefficient})$. Age and BMI spline terms jointly represent nonlinear associations and should not be interpreted individually.

Table S2.4: Subgroup discrimination in the primary and secondary outcome analyses.

Analysis	Subgroup	N	Events	Logistic	RF	XGBoost	LightGBM
Primary	Female	74,858	3,166	0.7158	0.7094	0.7168	0.7122
Primary	Male	58,976	4,863	0.6839	0.6797	0.6831	0.6795
Primary	Age < 60 years	80,629	2,977	0.7108	0.7060	0.7112	0.7065
Primary	Age ≥ 60 years	57,023	5,259	0.6457	0.6404	0.6465	0.6427
Secondary	Female	74,443	4,449	0.7076	0.7048	0.7090	0.7069
Secondary	Male	57,811	6,336	0.6899	0.6867	0.6904	0.6874
Secondary	Age < 60 years	79,516	3,873	0.6979	0.6931	0.6993	0.6949
Secondary	Age ≥ 60 years	56,582	7,232	0.6379	0.6365	0.6399	0.6380

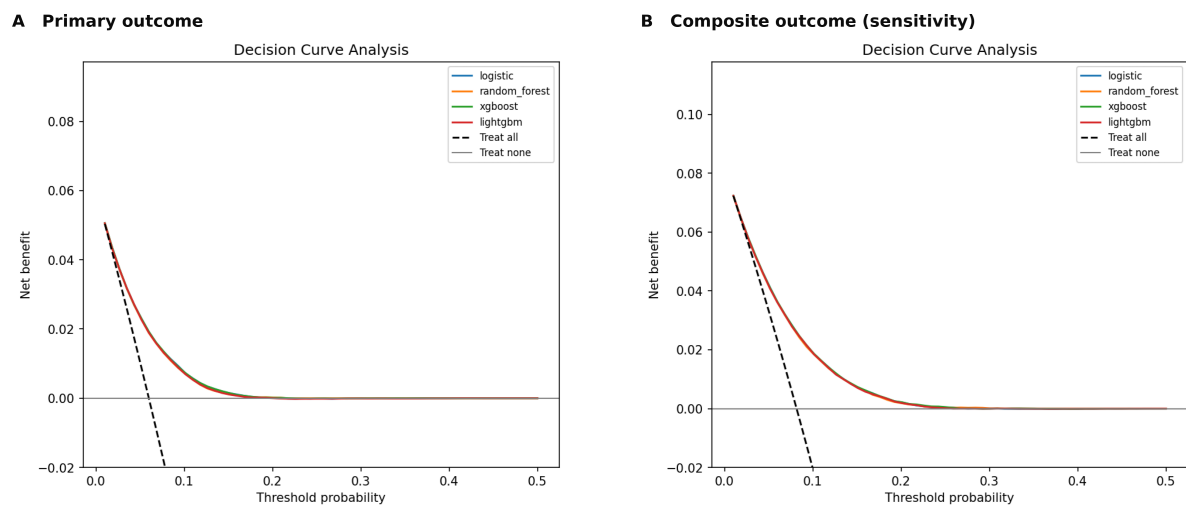


Figure S2.4: Decision-curve analysis for the four prediction models in the primary CVD outcome (A) and secondary outcome (B). Net benefit is shown across threshold probabilities, with treat-all and treat-none strategies as references.

S3. SUPPLEMENTARY RESULTS — CAUSAL MACHINE LEARNING

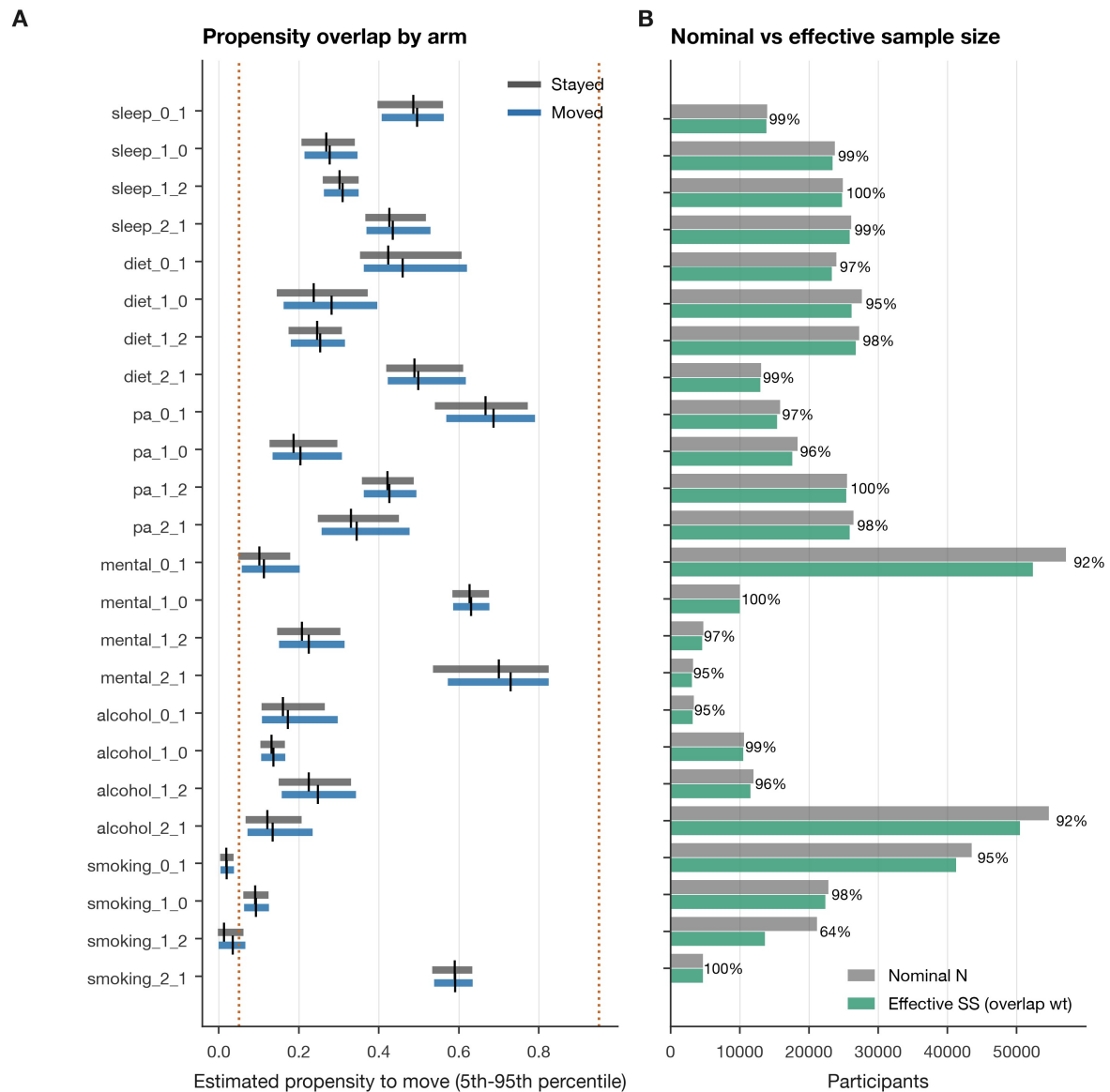


Figure S3.1: Overlap diagnostics for the 24 lifestyle transitions. (A) Propensity-score distributions in the stayed and moved arms, shown as 5th–95th percentile ranges with medians; dotted lines mark 0.05 and 0.95. (B) Nominal and overlap-weighted effective sample sizes; percentages indicate effective-to-nominal sample-size ratios.

S3.1 Baseline causal reference model results in detail

885 Five of the six domains behaved as expected, with a healthier baseline level associated with lower CVD risk. The largest differences were for smoking (never versus ever, -1.72 pp, 95% CI -1.86 to -1.58) and mental health (low versus higher symptom burden, -1.53 pp, -1.71 to -1.35), followed by sleep (-0.74 pp, -0.89 to -0.59), diet (-0.28

pp, -0.43 to -0.14) and physical activity (-0.22 pp, -0.36 to -0.08). Alcohol was the single exception. Participants in the lower-intake category had 1.35 pp higher risk ($+1.18$ to $+1.52$), the same paradoxical direction as the FDR-supported alcohol 2 $\rightarrow$ 1 transition. All six intervals excluded zero, reflecting the precision available in a cohort six times larger than the causal cohort rather than larger effects, and these estimates describe standing lifestyle level and are not comparable in magnitude with the transition ATEs. The CATE distributions were narrow but, unlike the transition forests, consistently signed. The alcohol distribution lay entirely above zero (interquartile range 0.86 pp) and the smoking and mental-health distributions entirely below it, consistent with a uniform level effect rather than usable individual-level heterogeneity (Table S3.1, Figure S3.2).

Table S3.1: Baseline-reference causal-forest estimates comparing healthy with less-healthy baseline lifestyle levels.

Domain	Healthy baseline			Less-healthy baseline			Adjusted effect	
	<i>N</i>	Events	Risk (%)	<i>N</i>	Events	Risk (%)	ATE (pp)	95% CI (pp)
Sleep	151,614	7,363	4.86	306,702	20,021	6.53	-0.74	$[-0.89, -0.59]$
Smoking	255,898	12,150	4.75	202,418	15,234	7.53	-1.72	$[-1.86, -1.58]$
Alcohol	136,891	8,997	6.57	321,425	18,387	5.72	$+1.35$	$[+1.18, +1.52]$
Diet	302,206	16,809	5.56	156,110	10,575	6.77	-0.28	$[-0.43, -0.14]$
Physical activity	167,038	9,534	5.71	291,278	17,850	6.13	-0.22	$[-0.36, -0.08]$
Mental health	345,323	19,636	5.69	112,993	7,748	6.86	-1.53	$[-1.71, -1.35]$

Note: ATE, average treatment effect; CI, confidence interval; pp, percentage points. Negative estimates indicate lower CVD risk under the healthy baseline level.

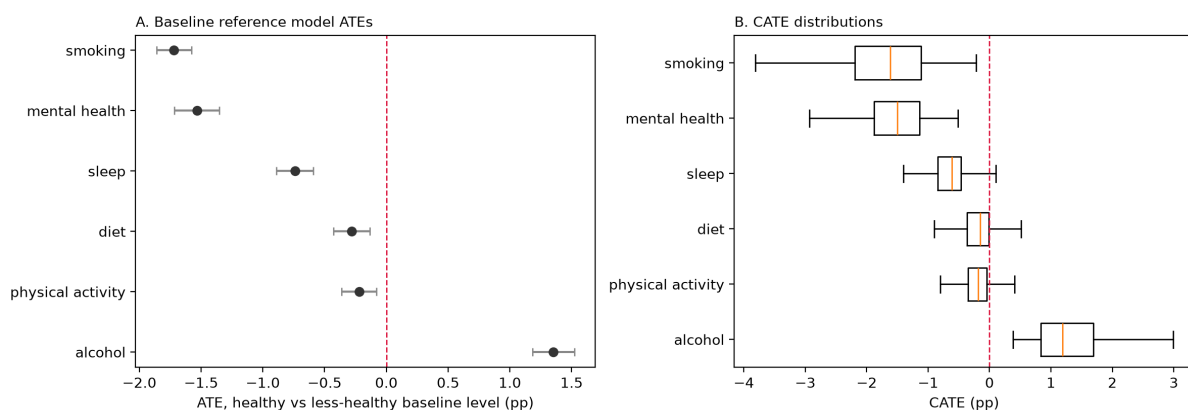


Figure S3.2: Baseline-reference causal-forest results comparing healthy with less-healthy lifestyle levels. (A) ATEs with 95% CI. (B) Distributions of individual CATEs. Dashed lines indicate no effect; effects are expressed in pp.

S3.2 Effect Heterogeneity analyses in detail

900 Variable-importance rankings are reported in Figure S3.4 as descriptive split-frequency measures, dominated by BMI and age, which is consistent with these variables predicting baseline risk rather than modifying treatment effects.

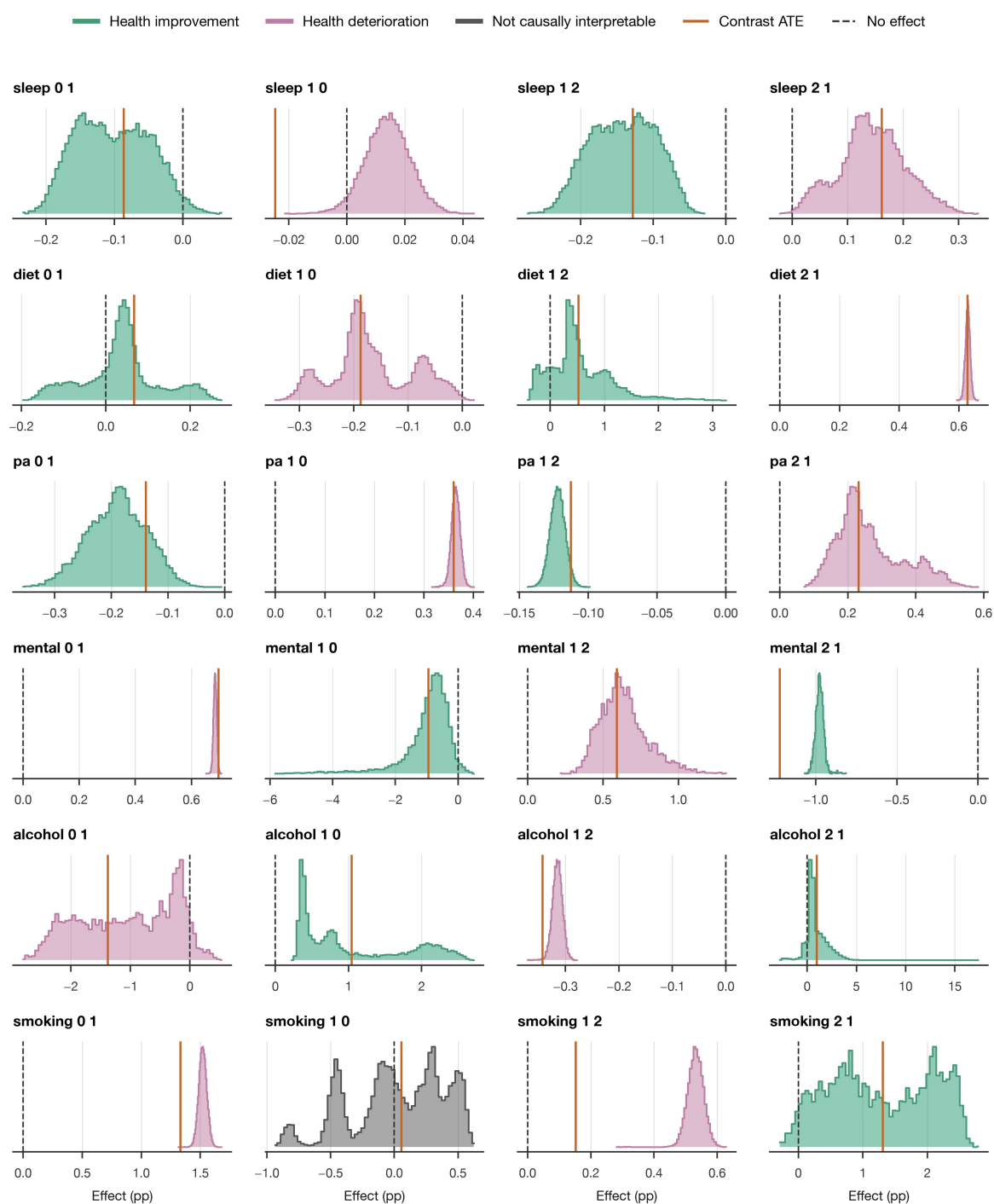


Figure S3.3: Distributions of individual conditional average treatment effects for the 24 lifestyle transitions. Effects are expressed in pp; orange lines show contrast-specific ATEs and dashed lines indicate no effect. Colours distinguish health improvement, deterioration and non-interpretable transitions.

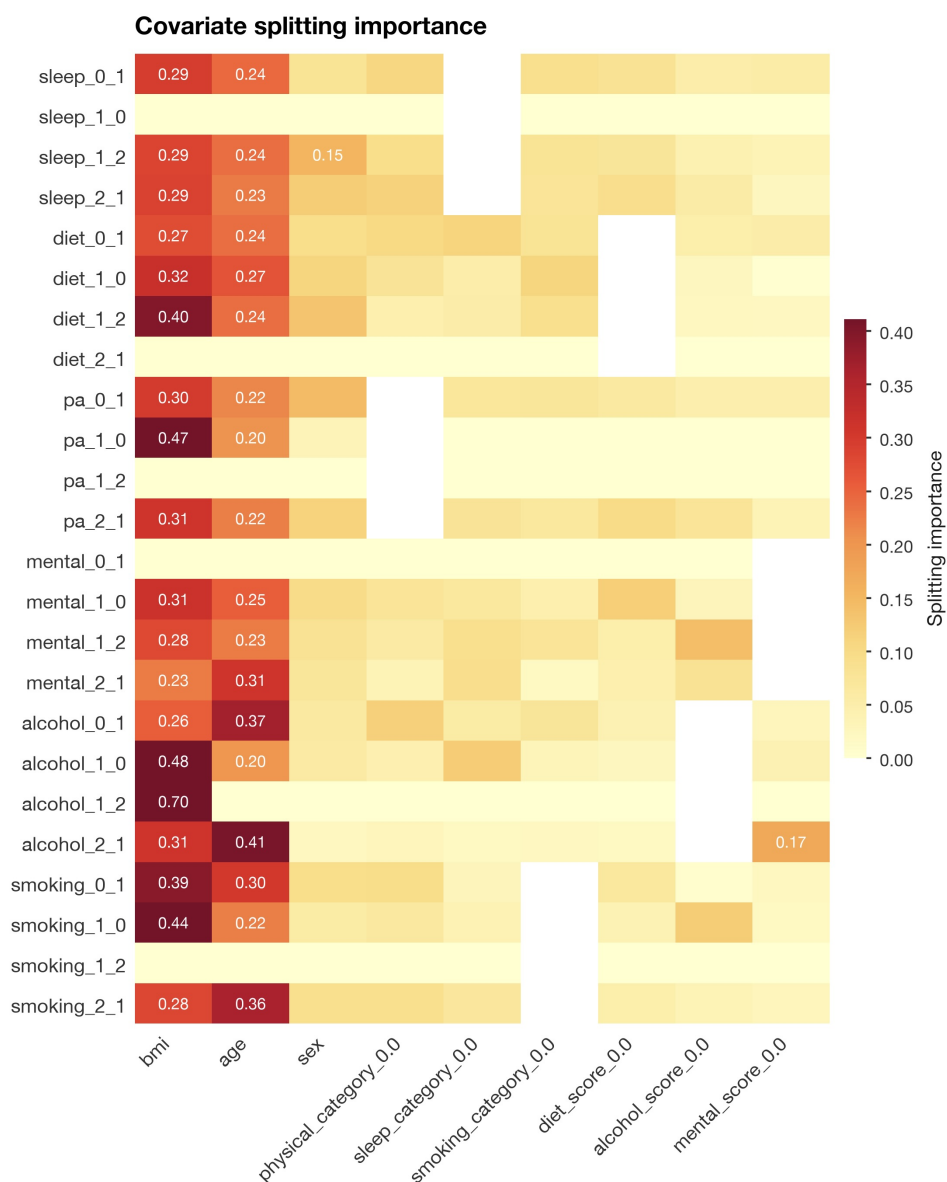


Figure S3.4: Causal-forest covariate splitting importance across the 24 lifestyle transitions. Darker cells indicate greater importance of a covariate in partitioning treatment-effect heterogeneity.

S3.3 Subgroup analyses in detail

Figure S3.5 reports the full subgroup ATE estimates by sex, age and BMI for every
905 transition, with both Wald and bootstrap difference tests across the 72-test family. The
largest nominal differences were the alcohol 2→1 age contrast (+2.16 pp for age ≥ 60
versus < 60 , bootstrap 95% CI +0.82 to +3.63), the alcohol 1→0 BMI contrast (+4.25
pp for BMI ≥ 30 versus < 25), and the alcohol 0→1 age contrast (−5.65 pp). The
remaining nominal contrasts (diet 2→1 and 1→2, physical activity 2→1) were smaller.
910 None survived FDR correction, and the two inference procedures agreed almost exactly,
indicating that the Wald standard errors were not materially understated.

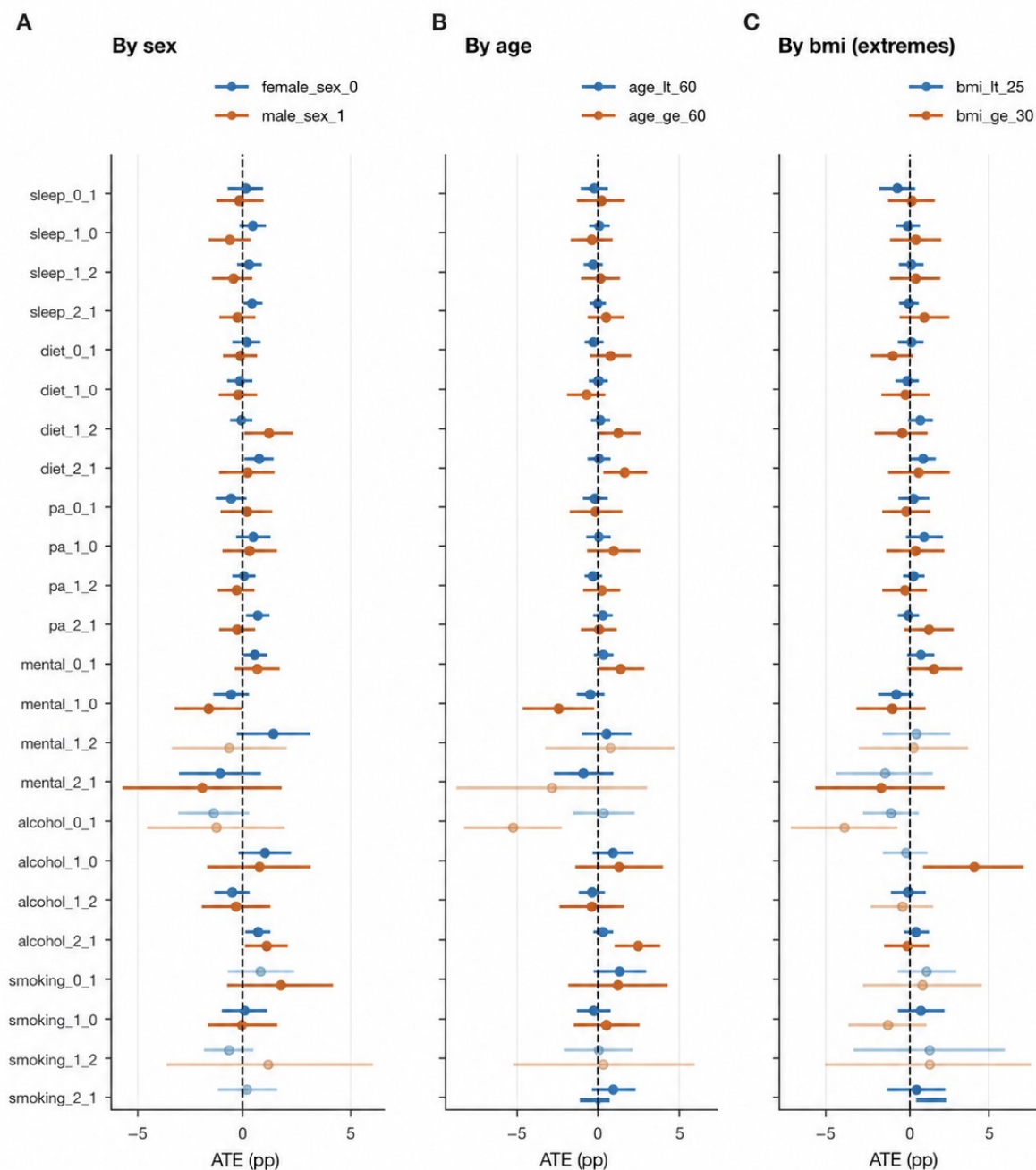


Figure S3.5: Subgroup-specific ATEs for the 24 lifestyle transitions by sex (A), age (B) and BMI extremes (C). Points show estimates and horizontal lines show 95% CI; dashed lines indicate no effect. Effects are expressed in pp.

S3.4 Combined-variable analysis in detail

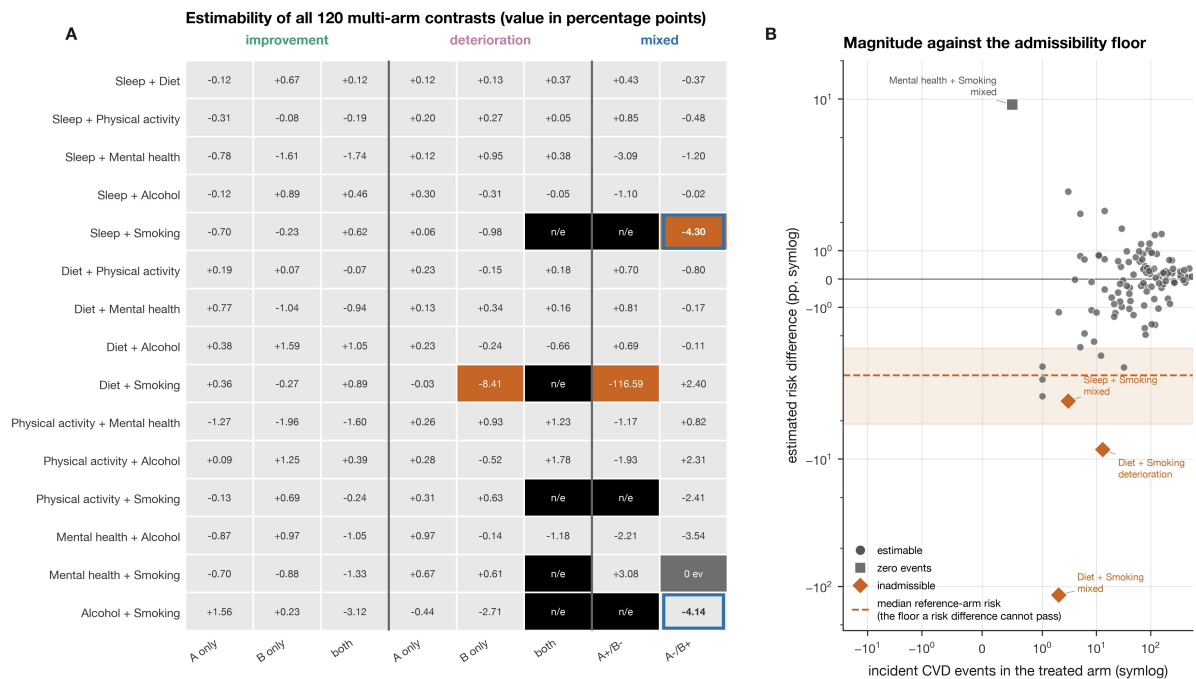


Figure S3.6: Estimability audit of 120 multi-arm contrasts. (A) Estimated risk differences by lifestyle pair and change scenario; black cells indicate non-estimable contrasts or zero events. (B) Effect magnitude against treated-arm events and the admissibility floor; orange diamonds identify inadmissible estimates.

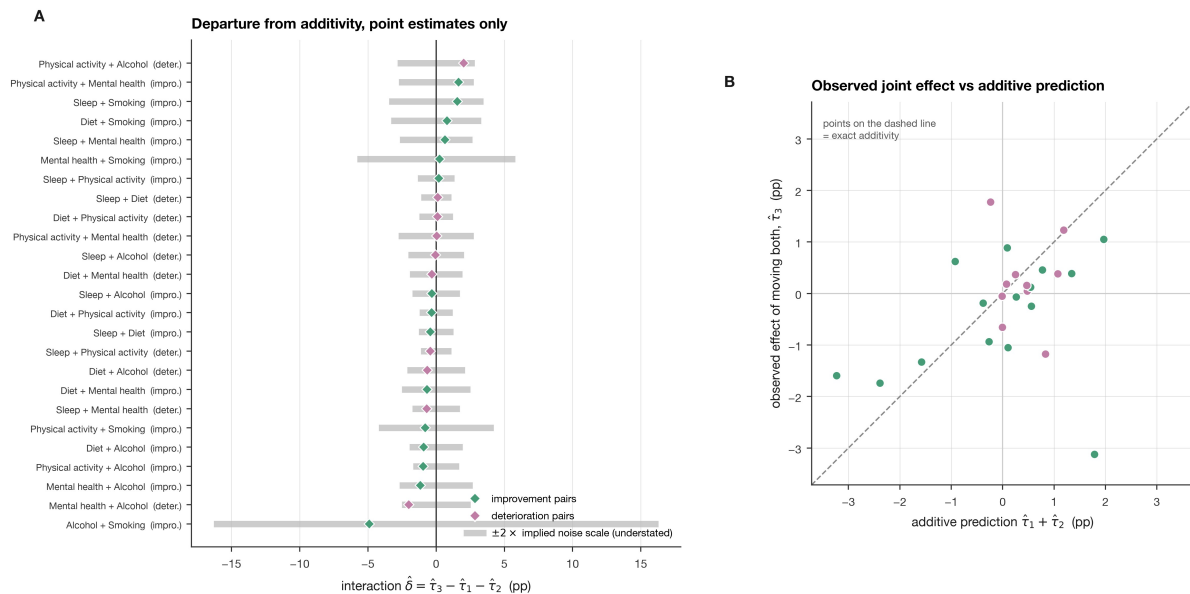


Figure S3.7: Assessment of additivity for joint lifestyle changes. (A) Departure of the observed joint effect from the sum of the single-change effects; zero indicates additivity. Grey bars show an approximate noise scale. (B) Observed joint effects versus additive predictions; the dashed line indicates exact additivity. Effects are expressed in pp.

S3.5 Complete sensitivity results

Figure S3.8 reports the per-estimand results with the retained fraction under each trimming window (0.01 to 0.99, 0.05 to 0.95, and 0.10 to 0.90) for all 24 contrasts. The estimate range across estimands was below 0.05 pp for 16 contrasts and below 0.35 pp for 23 of 24. The three sign changes are individually explained in the table notes. Smoking 1→2 is a thin-support contrast that trimming empties, retaining 6.3% of the sample at the intermediate window and none at the strictest. Smoking 1→0 is the not-causally-interpretable contrast. Sleep 1→0 has a magnitude of 0.02 pp, far below its own standard error.

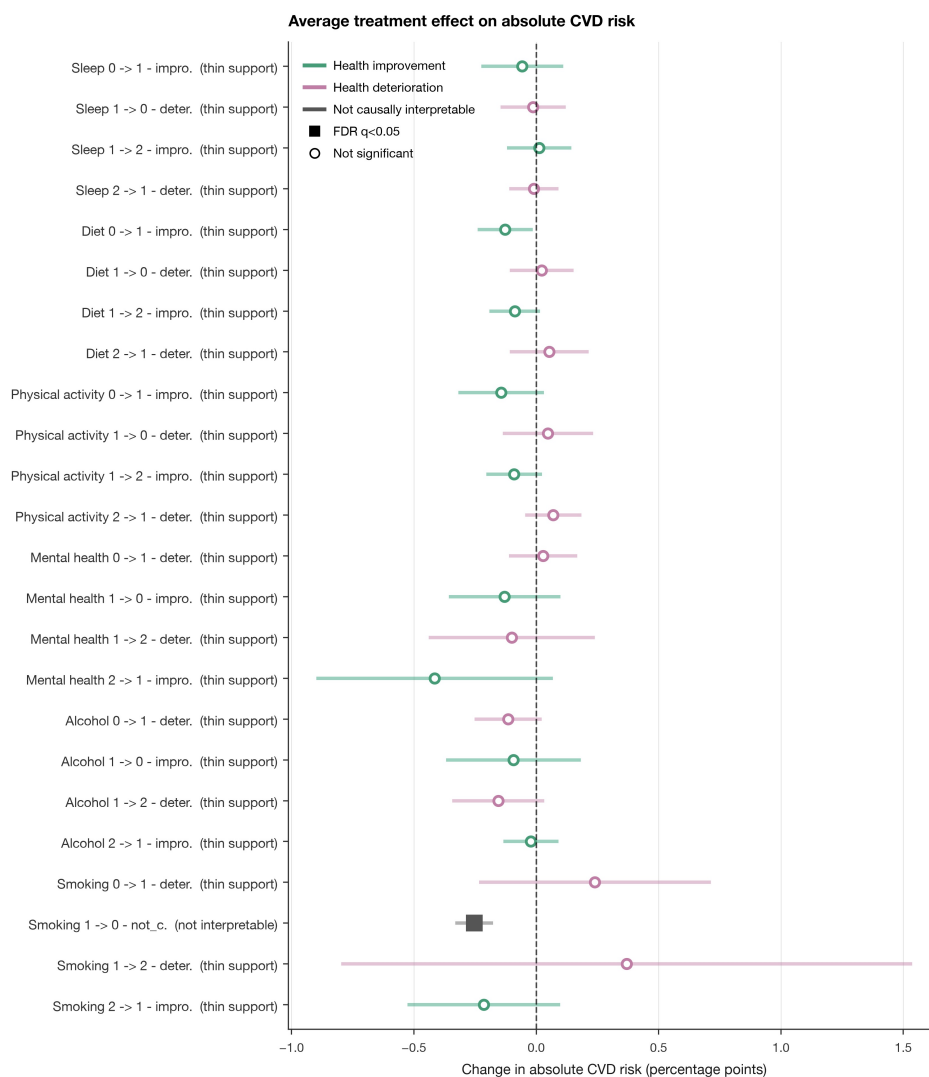


Figure S3.8: Average treatment effects for the 24 lifestyle transitions in the sensitivity analysis. Points and horizontal lines show risk differences and 95% CI in pp; the dashed line indicates no effect. Colours denote transition direction, and filled squares indicate FDR-supported estimates.

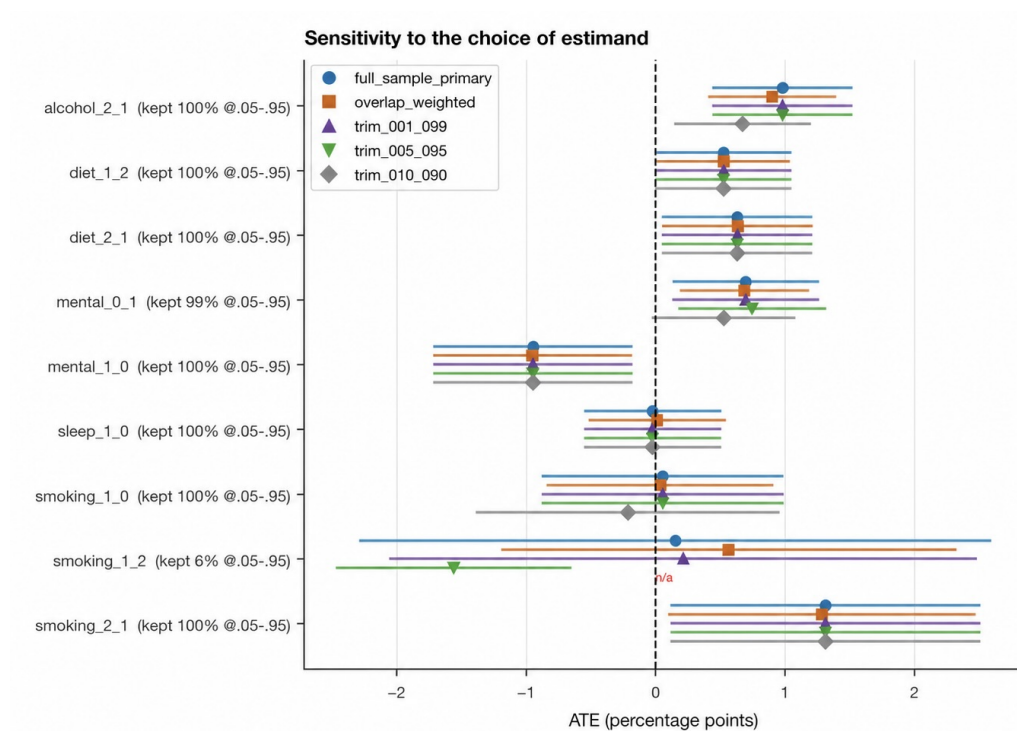


Figure S3.9: Sensitivity of selected ATEs to the estimand. Full-sample, overlap-weighted and propensity-score-trimmed estimates are shown with 95% CI; the dashed line indicates no effect.

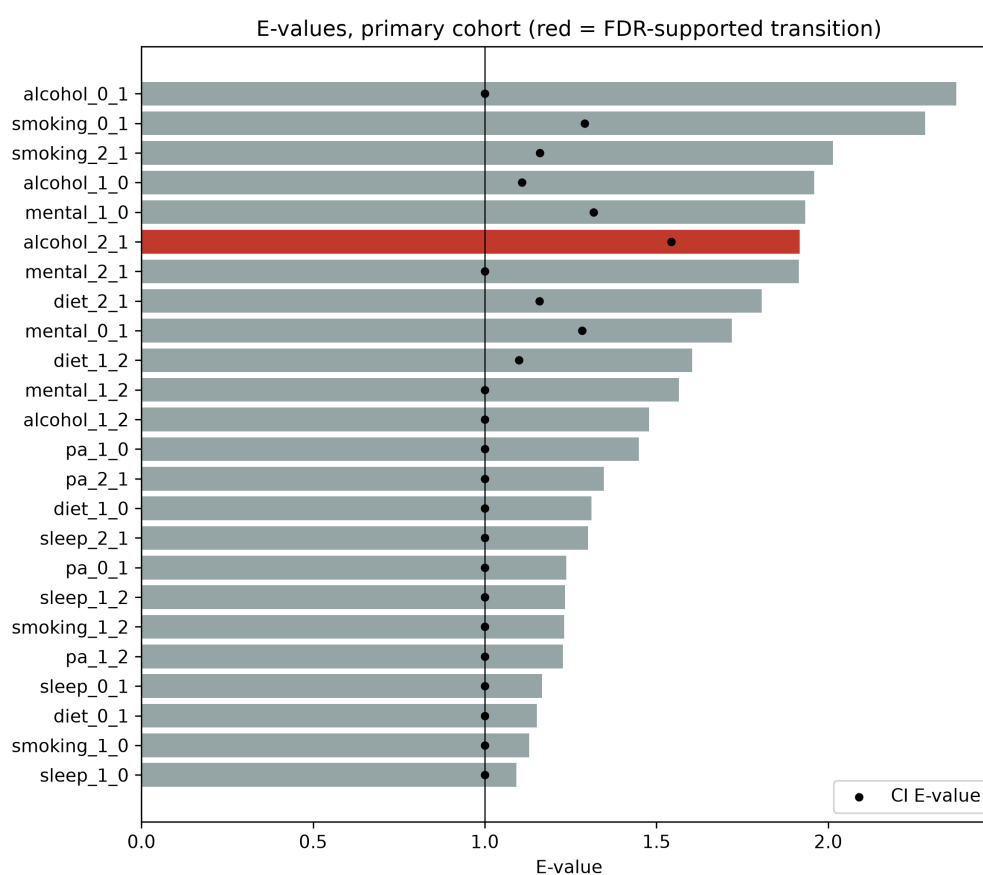


Figure S3.10: E-values for unmeasured confounding across the 24 lifestyle transitions. Bars show point-estimate E-values and black points show confidence-limit E-values; the vertical line marks the null value of 1. The FDR-supported transition is shown in red.

Table S3.2: Average treatment effects in the NHEFS external validation analysis

Contrast	Health direction	<i>N</i>	Events	ATE, pp (95% CI)	<i>p</i> -value	FDR-adjusted <i>p</i>	Assessment
pa_0_1	Improvement	1,969	223	−2.289 (−5.505, 0.927)	0.163	0.448	No evidence of an effect (thin support)
pa_1_0	Deterioration	3,069	233	−0.069 (−1.990, 1.852)	0.944	0.944	No evidence of an effect
pa_1_2	Improvement	1,377	81	−1.650 (−3.792, 0.493)	0.131	0.448	No evidence of an effect (thin support)
pa_2_1	Deterioration	676	41	0.829 (−2.486, 4.143)	0.624	0.769	No evidence of an effect (thin support)
alcohol_0_1	Deterioration	2,662	317	−2.135 (−4.995, 0.725)	0.143	0.448	No evidence of an effect
alcohol_1_0	Improvement	3,055	196	2.410 (−0.261, 5.080)	0.077	0.448	No evidence of an effect (thin support)
alcohol_1_2	Deterioration	2,835	161	−1.026 (−4.492, 2.441)	0.562	0.769	No evidence of an effect (thin support)
alcohol_2_1	Improvement	978	50	0.559 (−2.170, 3.287)	0.688	0.769	No evidence of an effect (thin support)
smoking_0_1	Deterioration	3,207	258	−1.994 (−6.763, 2.775)	0.413	0.769	No evidence of an effect (thin support)
smoking_1_2	Deterioration	1,776	177	2.641 (−4.564, 9.845)	0.473	0.769	No evidence of an effect (thin support)
smoking_2_1	Improvement	1,939	144	−0.525 (−3.183, 2.133)	0.699	0.769	No evidence of an effect (thin support)

Notes: ATE denotes the average treatment effect; pp denotes percentage points; CI denotes confidence interval. FDR-adjusted *p*-values were calculated within the prespecified family of external-validation contrasts. “Thin support” indicates limited participant or event support for the corresponding transition. None of the confidence intervals excluded zero.

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